Keldysh-Ionization-Based FDTD Modeling and Bayesian Optimization of Laser-Induced Damage Thresholds in Hybrid Gratings

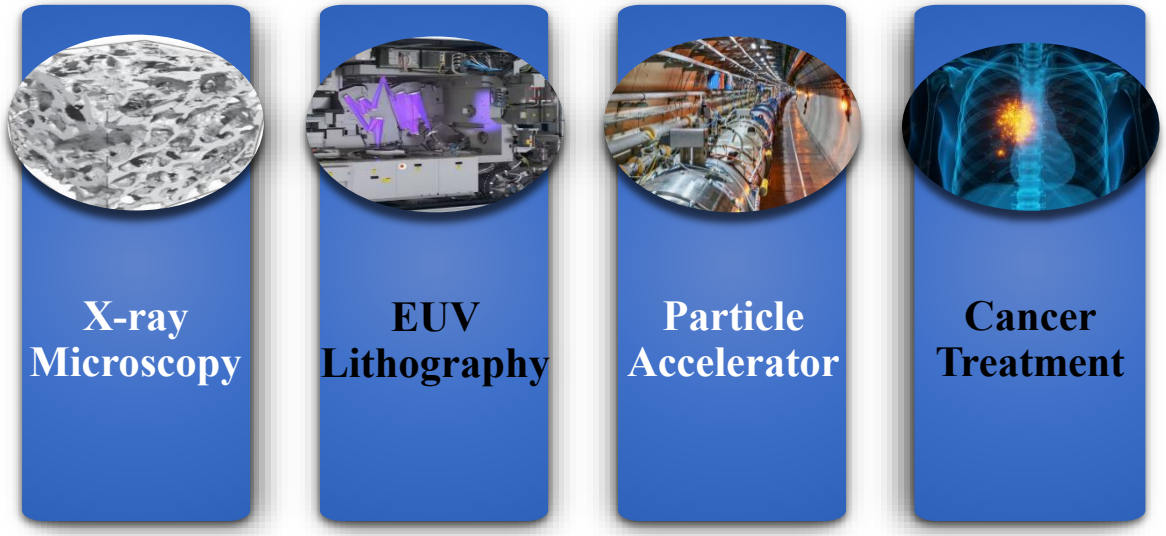
Ziyao Su[1], Johnny Eschbach[2,3], Enam A. Chowdhury[1,2,3]

[1] *The Ohio State University, Department of Materials Science and Engineering, Columbus, OH 43210*
[2] *Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH 43210, USA*
[3] *Department of Physics, The Ohio State University, Columbus, OH 43210, USA, OH 43210*

Acknowledgements: **This work was supported by the United States Air Force Office of Scientific Research (FA9550-25-1-0297) and the DOE ARDAP (DE-SC0025577) and used resources from the Ohio Supercomputer Center (OSC).**

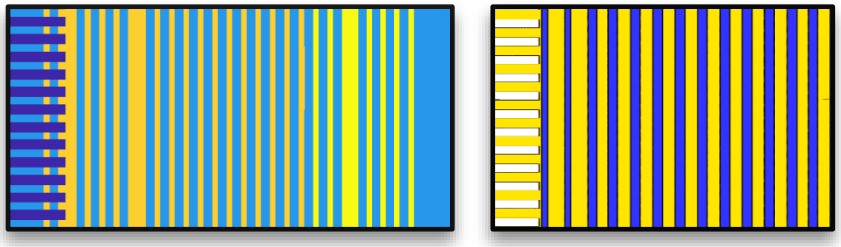
Motivation

Real-world laser-driven accelerators, x-ray and ion sources need **powerful, efficient ultrashort short-pulse lasers** with **high peak and average power**.



Failure of optical components, such as **compression grating**, is a key impediment of existing CPA-based ultrafast, high-energy laser systems [1,2]. Therefore, it is crucial to increase the **Laser Induced Damage Threshold (LIDT)**.

- **Multilayer dielectric (MLD) gratings**: Low absorption and high reflectivity, but broadband optimization increases design and fabrication complexity.



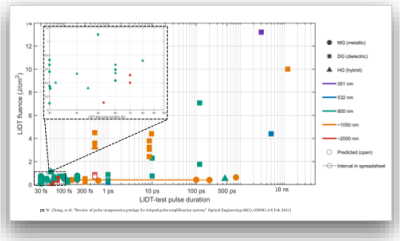
~20 to 50 layers

$SiO_2/HfO_2/Ta_2O_5$
48 layers at 800 nm [5]

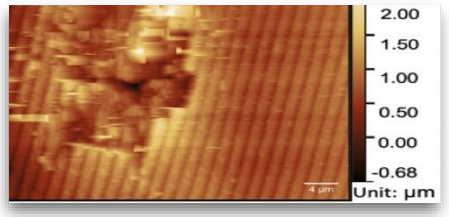
SiO_2/HfO_2
27 layers at 1950 nm [6,7]

- **Hybrid gratings**: Offer a promising balance of broad bandwidth, low loss, high damage resistance and reduced layers for femtosecond PW lasers.

~10 layers

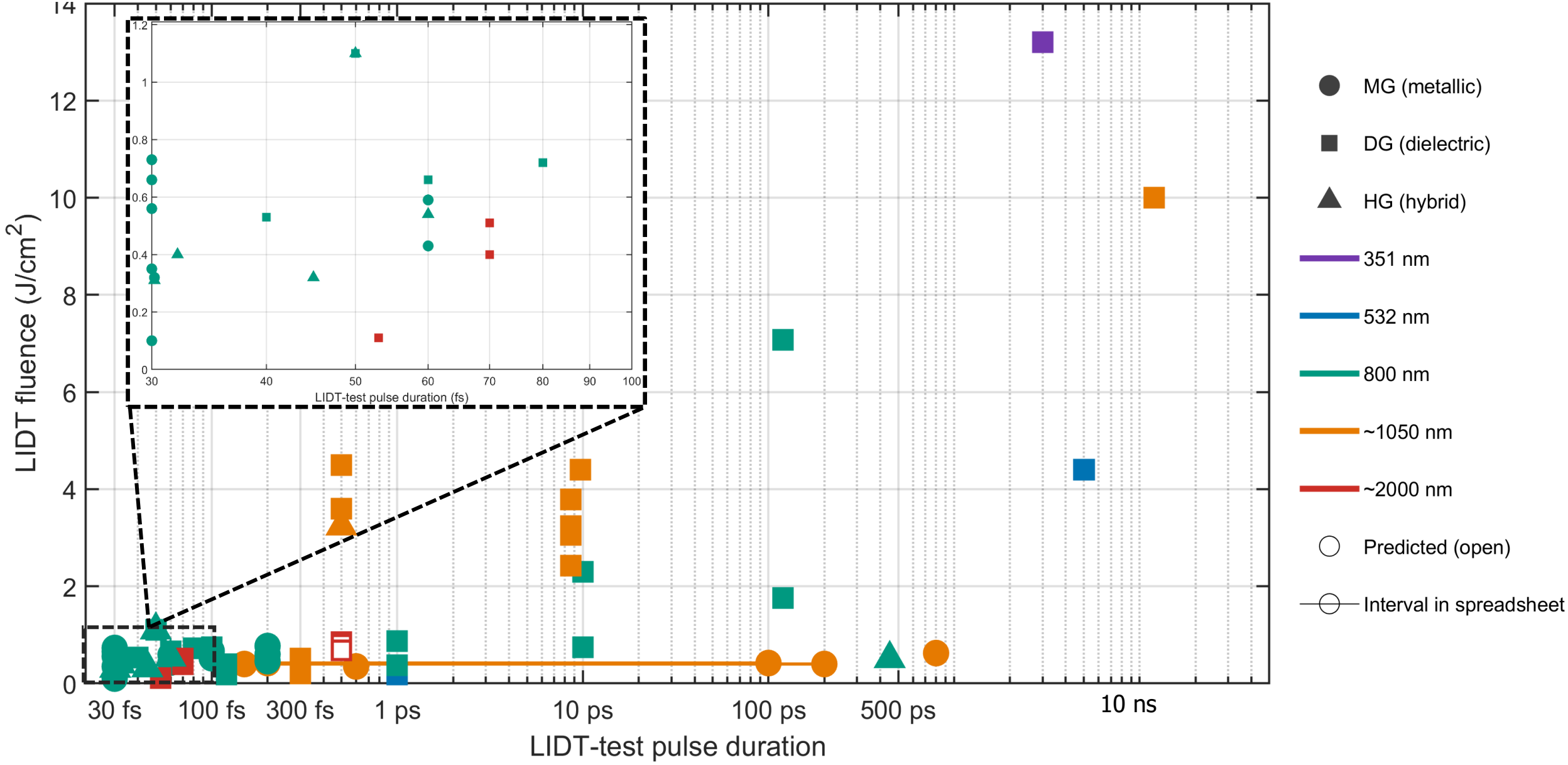


Grating LIDT development in the past three decades [3].



Typical LID observed in the multilayer dielectric gratings [4].

[1] M. F. Kling, et al., Basic Research Needs Workshop on Laser Technology Report of the Department of Energy Office of Science Workshop August 15-17, 2023 (U.S. Department of Energy Office of Science, January 5, 2024), iii, 4, 44, 49-51, 54, etc.
[2] L Kiani., et al., "High Average Power Ultrafast Laser Technologies for Driving Future Advanced Accelerators" (technical report), Journal of Instrumentation 18, no. 08 (2023): 9
[3] W. Zhang, et al. "Review of pulse compression gratings for chirped pulse amplification system," Optical Engineering 60(2), 020902 (10 Feb 2021)
[4] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)
[5] S. Zhang, Z. Su, C. S. Menoni, and E. A. Chowdhury, "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings," Opt. Lett., vol. 48, no. 5, pp. 1212–1215, Mar. 2023.
[6] Ziyao Su, et al., "Keldysh ionization based-FDTD modeling of laser-induced damage threshold of MLD-IBS compression gratings for petawatt 2 μm laser systems", Proc. SPIE PC13729, Laser-Induced Damage in Optical Materials 2025, PC1372907 (19 Dec 2025);
[7] M. Y. Noor, et al., "High Damage Threshold Multilayer Dielectric Grating Enabled for 2 μm High-Energy CPA Systems," Manuscript in preparation.



[3] W. Zhang, et al. "Review of pulse compression gratings for chirped pulse amplification system," Optical Engineering 60(2), 020902 (10 Feb 2021)

Our Efforts

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2021

2022

2023

2025

[4] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)

[8] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)

[5] S. Zhang, Z. Su, C. S. Menoni, and E. A. Chowdhury, "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings," Opt. Lett., vol. 48, no. 5, pp. 1212–1215, Mar. 2023.

[9] M. Y. Noor, et al., "Pulse duration dependent effects of ultrafast laser induced damage on a 1030 nm multi-layer dielectric mirror for high repetition rate, high average power laser systems," Opt. Mater. Express 15, 212-228 (2025)

Ultrashort pulse (< 150 fs) – Dielectric Coating Modeling

TOPIC 1

Ultrashort pulse (< 1 ps) – Hybrid Coating Modeling with TFSF Technique

2026

[10] Z. Su and E. A. Chowdhury, "Ultrafast laser interaction with dielectric–metal hybrid coatings: a two-dimensional finite-difference time-domain modeling study," High Power Laser Science and Engineering, vol. 14, p. e78, 2026.

[11] Z. Su and E. A. Chowdhury, "Pulse-duration scaling of ultrafast laser-induced damage threshold in hybrid gratings," Opt. Lett. 51, 4236-4239 (2026)

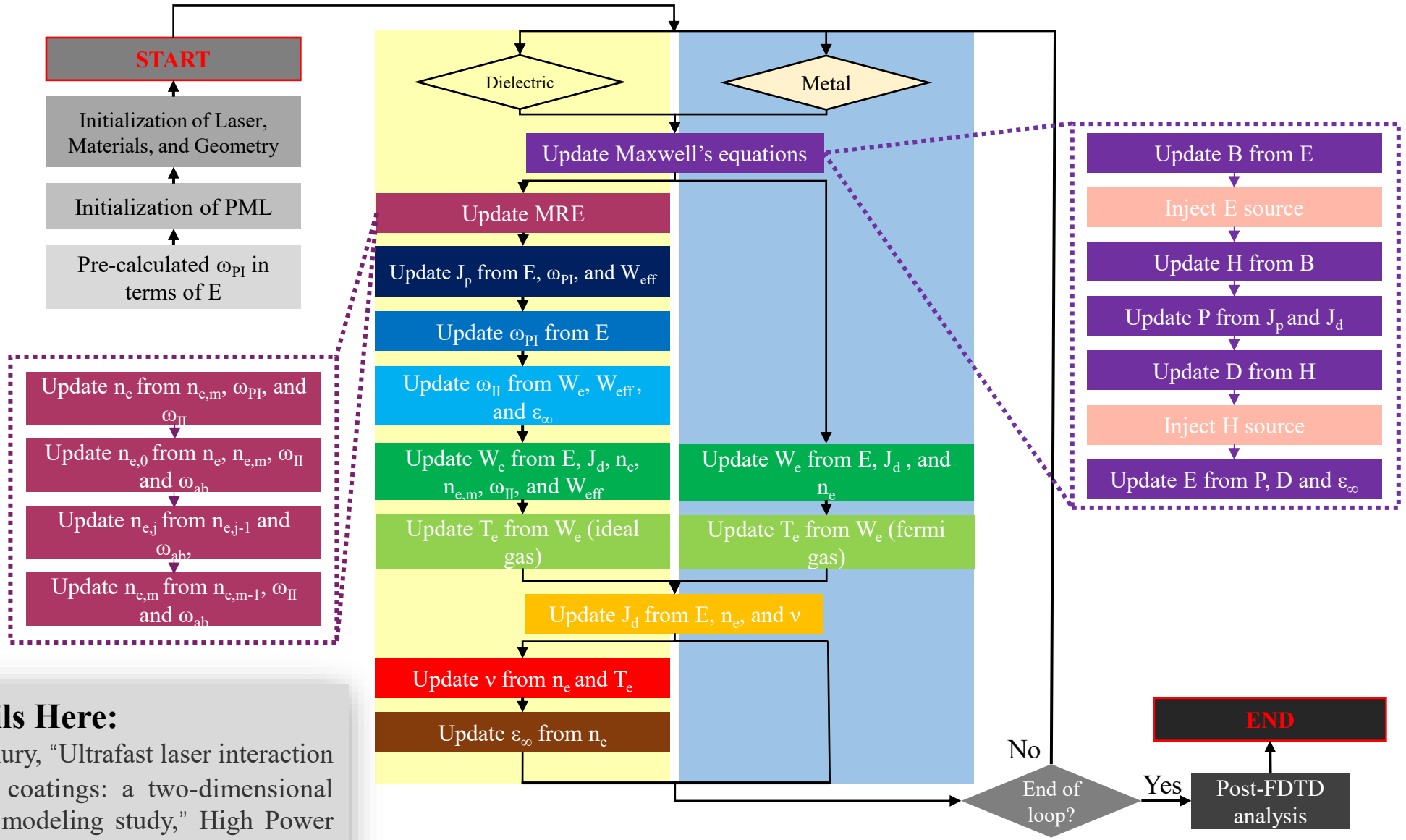
Z. Su, et al., "Machine Learning Damage-Aware Optimization of the Dielectric Compression Grating for Ultrafast High-Power Lasers," Manuscript in preparation.

Upcoming

TOPIC 2

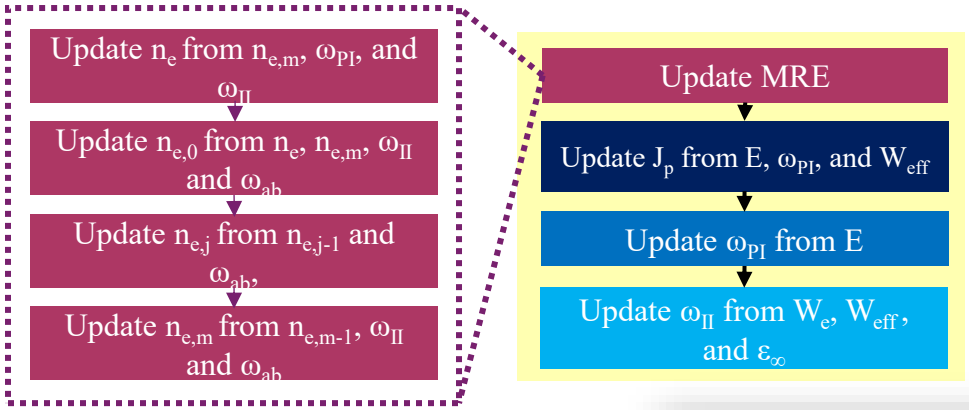
Machine Learning of Damage Aware Optimization Framework

Methods FDTD Framework

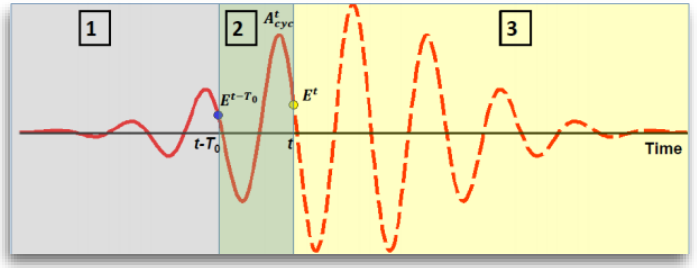


More Details Here:

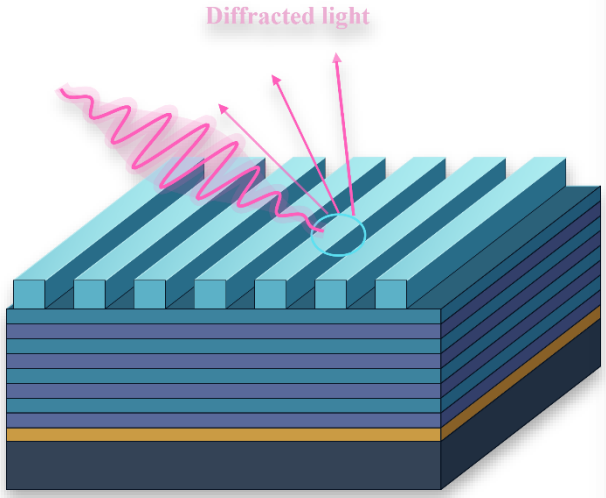
[10] Z. Su and E. A. Chowdhury, "Ultrafast laser interaction with dielectric–metal hybrid coatings: a two-dimensional finite-difference time-domain modeling study," High Power Laser Science and Engineering, vol. 14, p. e78, 2026.

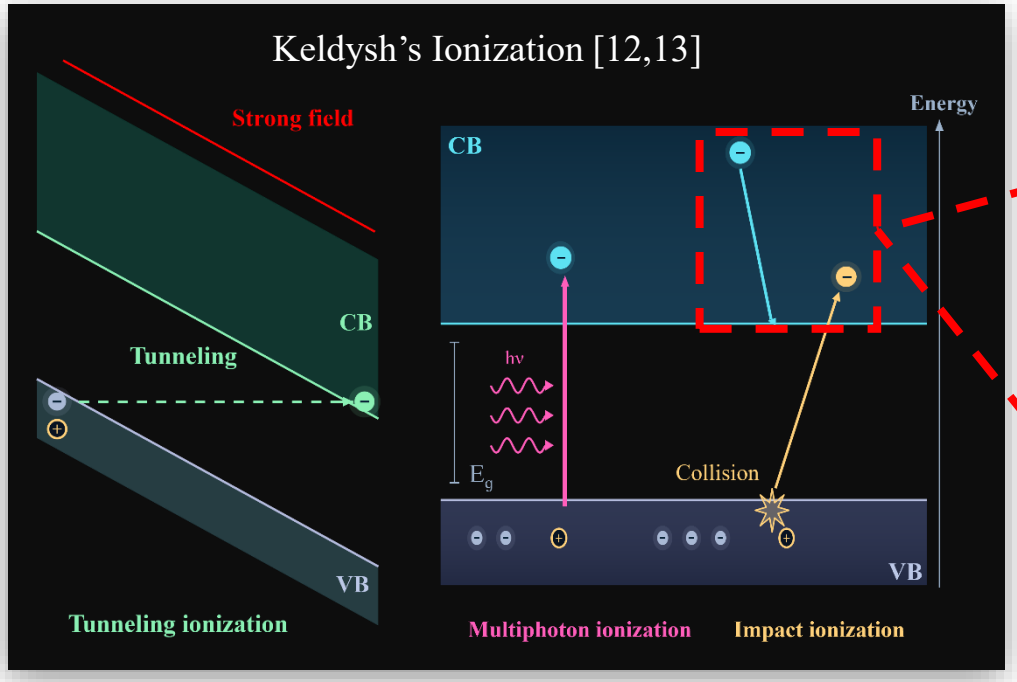


Cycle-averaged approach [8]



A sketch of time profile of electric field at a fixed space point





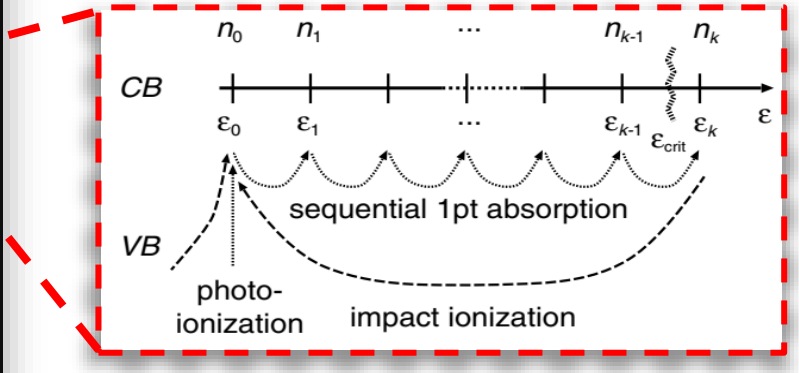


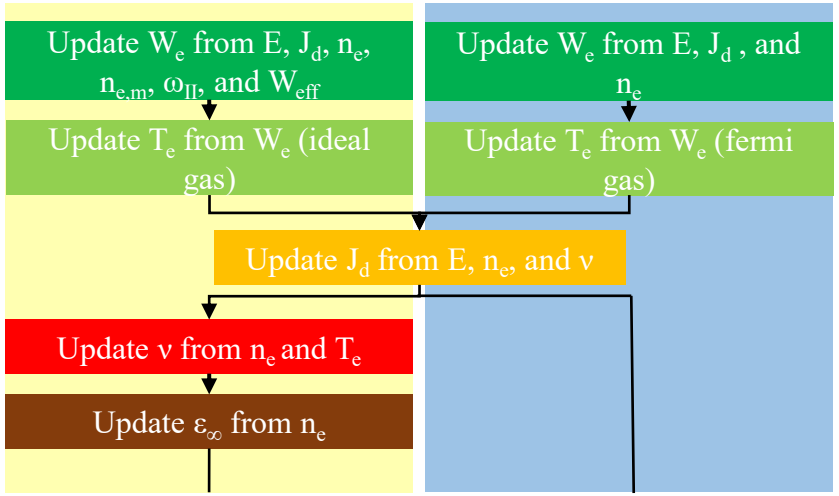
Illustration of the virtual energy ladders of free electrons in the conduction band of a dielectric [14].

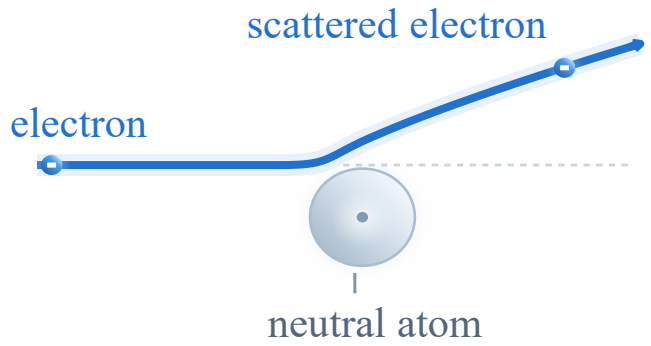
[8] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)
[12] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[13] L. V. Keldysh, "Kinetic theory of impact ionization in semiconductors," SOVIET PHYSICS JETP, vol. 37, no. 10, 1960.
[14] B. Rethfeld, "Unified Model for the Free-Electron Avalanche in Laser-Irradiated Dielectrics," Physical Review Letters 92, 187401 (2004).

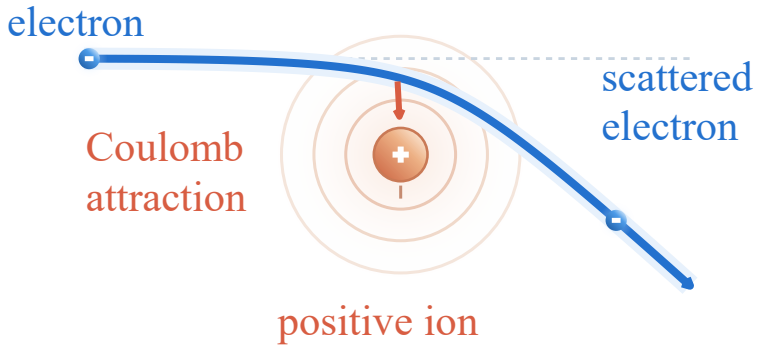
Methods

FDTD Framework | Collisions & Heating

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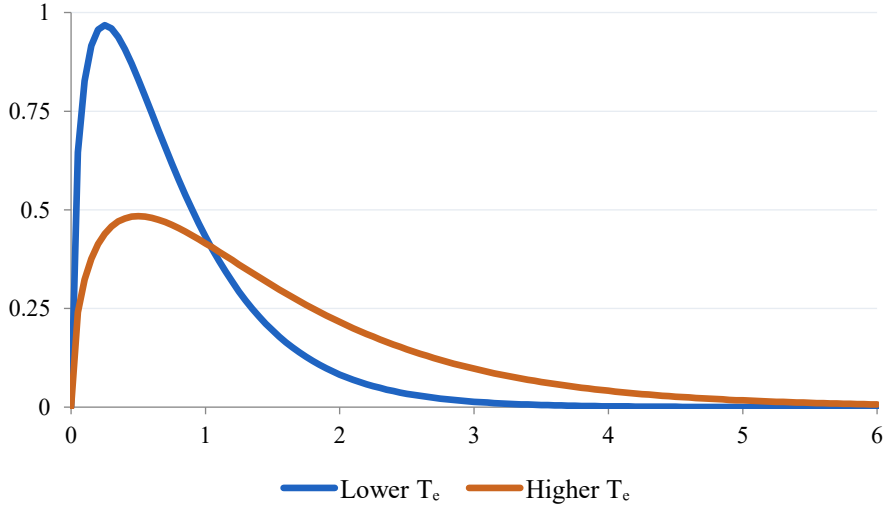




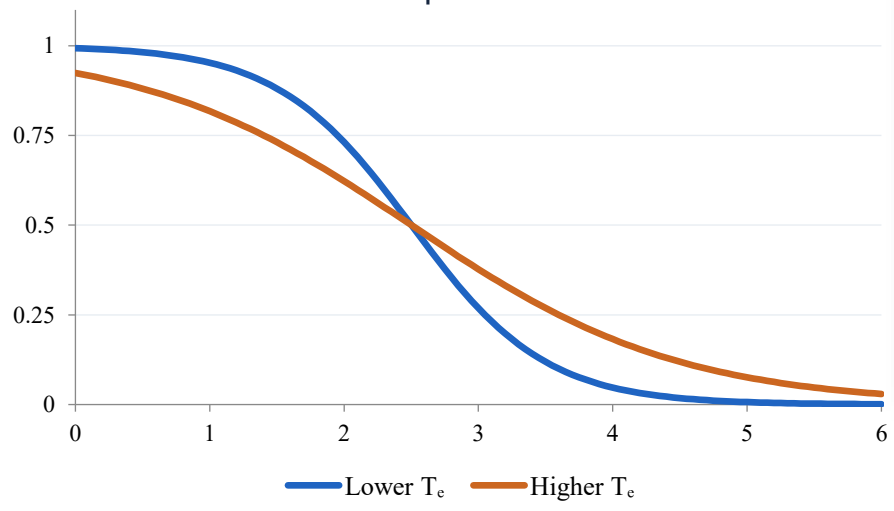


All collisions are globally non-linear, since electron density is transient.

Dielectric | Maxwell–Boltzmann



Metal | Fermi–Dirac



Limitation

The framework supports pulse durations from **10 fs to 1 ps** simulations based on these rapid thermalization assumptions.

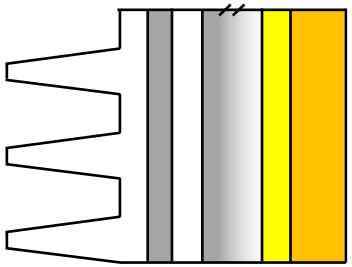
For pulse duration > ps, the lattice-electron energy exchange is not considered yet.

Simulation Set-up

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SiO_2
HfO_2
HfO_2\SiO_2 Pairs
Au
Substrate

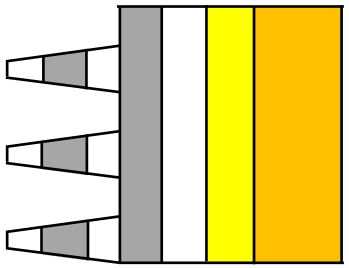
(a) Néauport *et al.* [15]



3 pairs

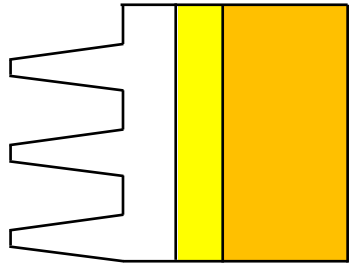
1057 nm
500 fs
AOI 77.2˚
s-pol
10 layers

(b) Guan *et al.* [16]



800 nm
45 fs
AOI 53˚
s-pol
6 layers

(c) Xu *et al.* [17]



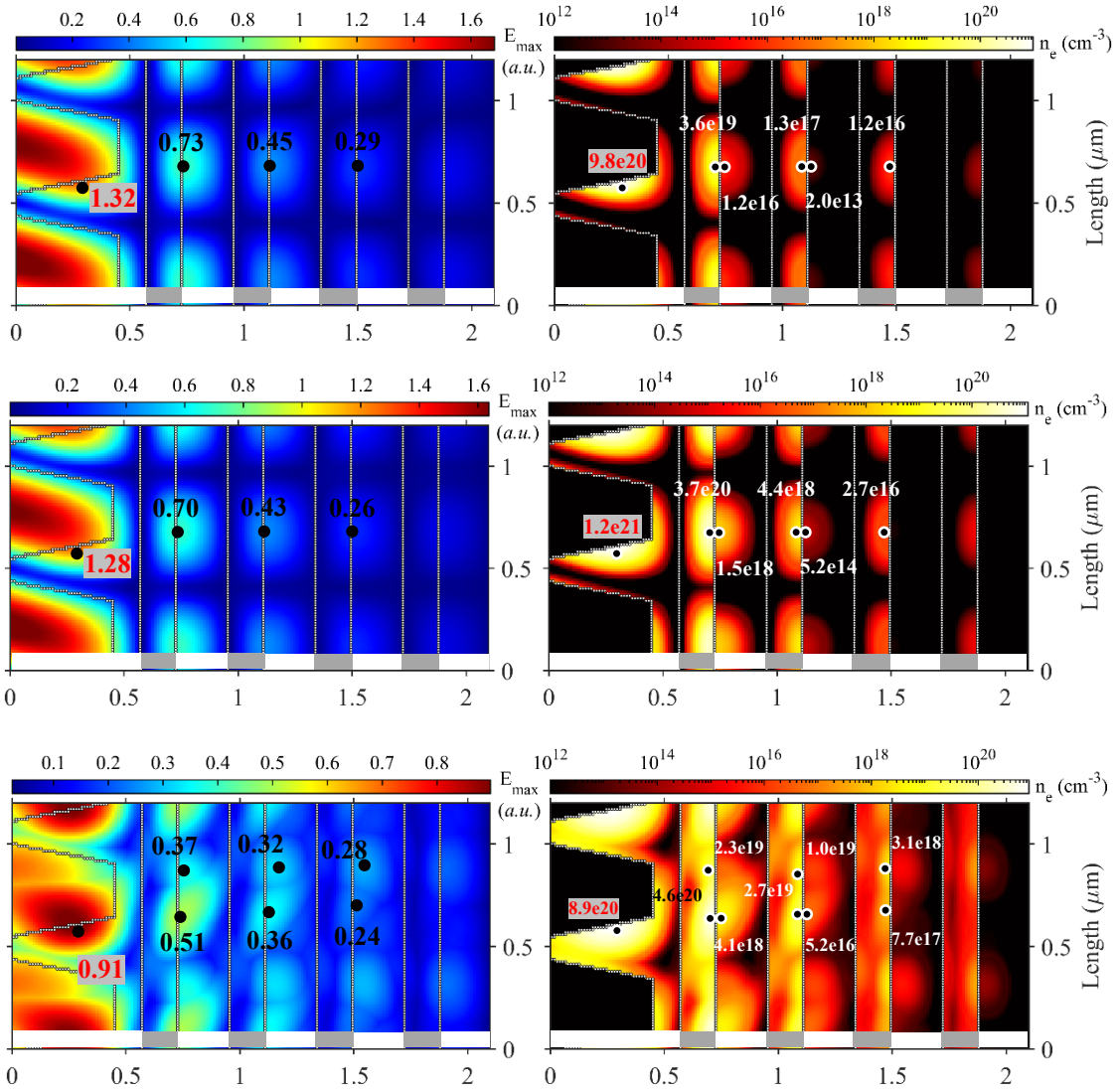
800 nm
30 fs
AOI 65˚
s-pol
2 layers

Design	Material	W_0 (eV)	ñ	m^*_e (a.u.)	m^*_h (a.u.)	ρ (g/cm³)
Néauport	SiO_2	9.0	1.45	0.60	8.00	2.20
	HfO_2	5.7	2.06	1.09	1.12	9.68
	Au	—	0.17 + i7.05	1.00	1.00	19.32
Guan	SiO_2	9.0	1.44	0.60	8.00	2.20
	HfO_2	5.7	1.96	1.09	1.12	9.68
	Au	—	0.18 + i5.00	1.00	1.00	19.32
Xu	SiO_2	8.5	1.44	0.50	8.00	2.20
	Au	—	0.18 + i5.00	1.00	1.00	19.32

More Details Here:

[11] Z. Su and E. A. Chowdhury, "Pulse-duration scaling of ultrafast laser-induced damage threshold in hybrid gratings," Opt. Lett. 51, 4236-4239 (2026)

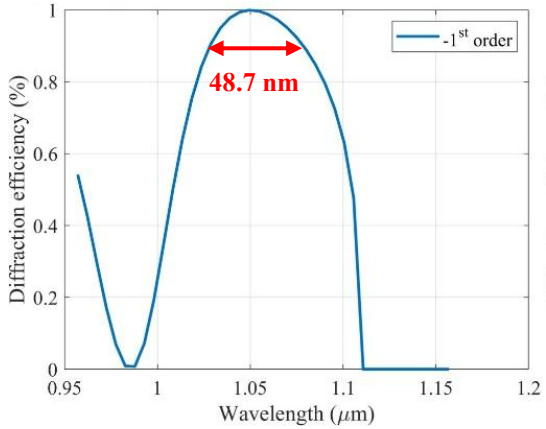
[15] J. Néauport, N. Bonod, S. Hocquet, et al., "Mixed metal dielectric gratings for pulse compression," Opt. Express 18, 23776–23783 (2010).
[16] H. Guan, H. Chen, J. Wu, et al., "High-efficiency, broad-bandwidth metal/multilayerdielectric gratings," Opt. Lett. 39, 170–173 (2014).
[17] J. Xu, X. Zou, J. Chen, et al., "Metal dielectric gratings with high femtosecond laser damage threshold of twice as much as that of traditional gold gratings," Opt. Lett. 44, 2871–2874 (2019).

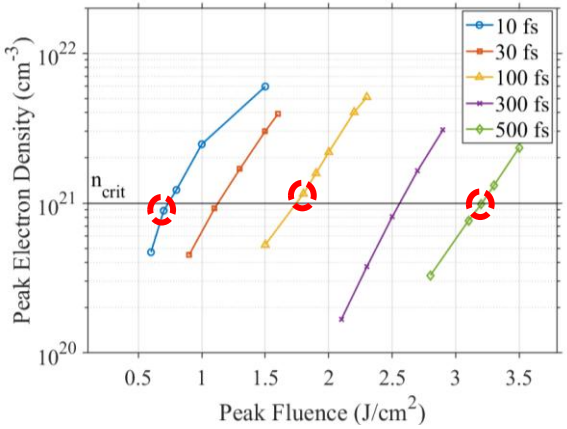


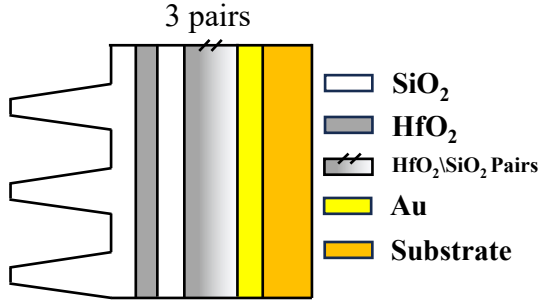
500fs
3.2 J/cm²

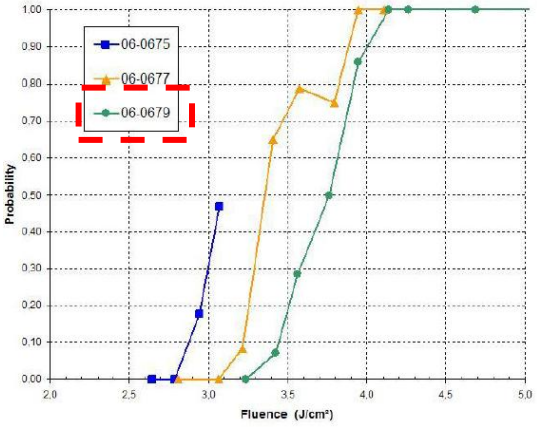
100fs
1.8 J/cm²

10fs
0.7 J/cm²







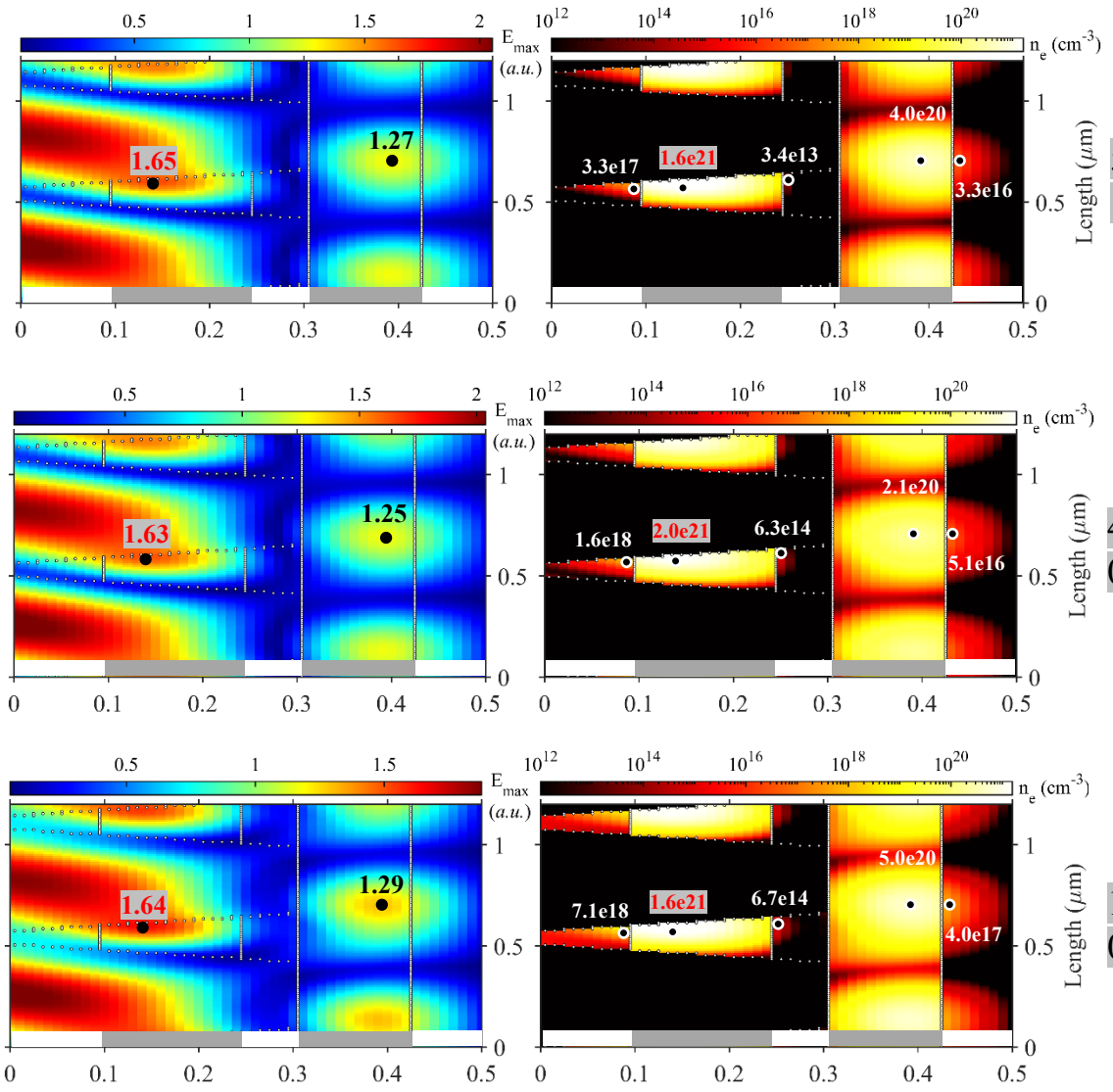


Damage probability of MMLD gratings. 1057nm, 500fs, TE polarization, at the incidence of 77.2°. Fluences are given in beam normal. Tested LIDT for the selected design is ~3.24 J/cm². [15]

[15] J. Néauport, N. Bonod, S. Hocquet, et al., "Mixed metal dielectric gratings for pulse compression," Opt. Express 18, 23776–23783 (2010).

Results Guan

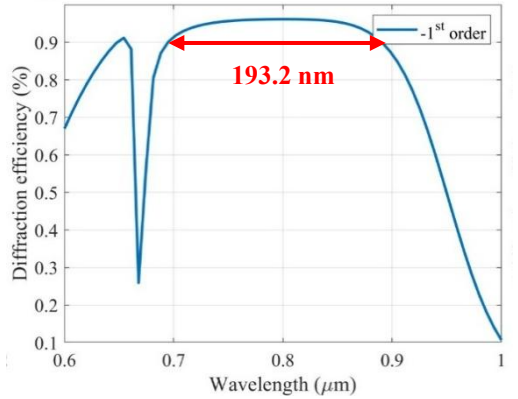
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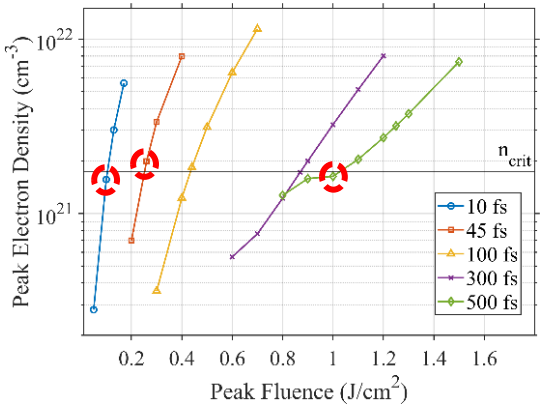


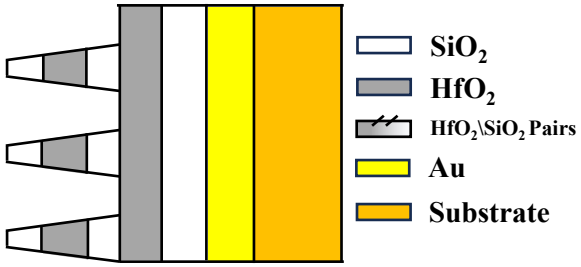
500fs
1.0 J/cm^2

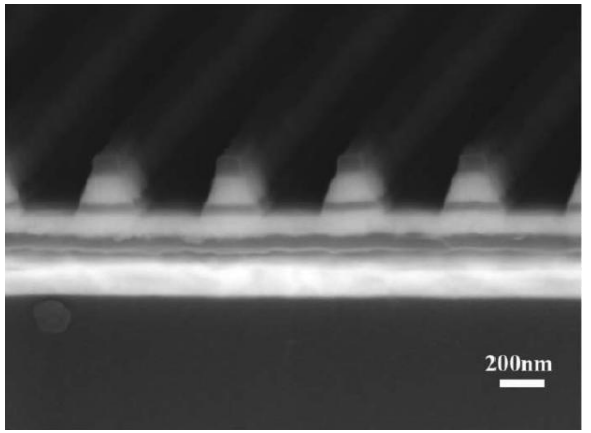
45fs
0.26 J/cm^2

10fs
0.1J/cm^2







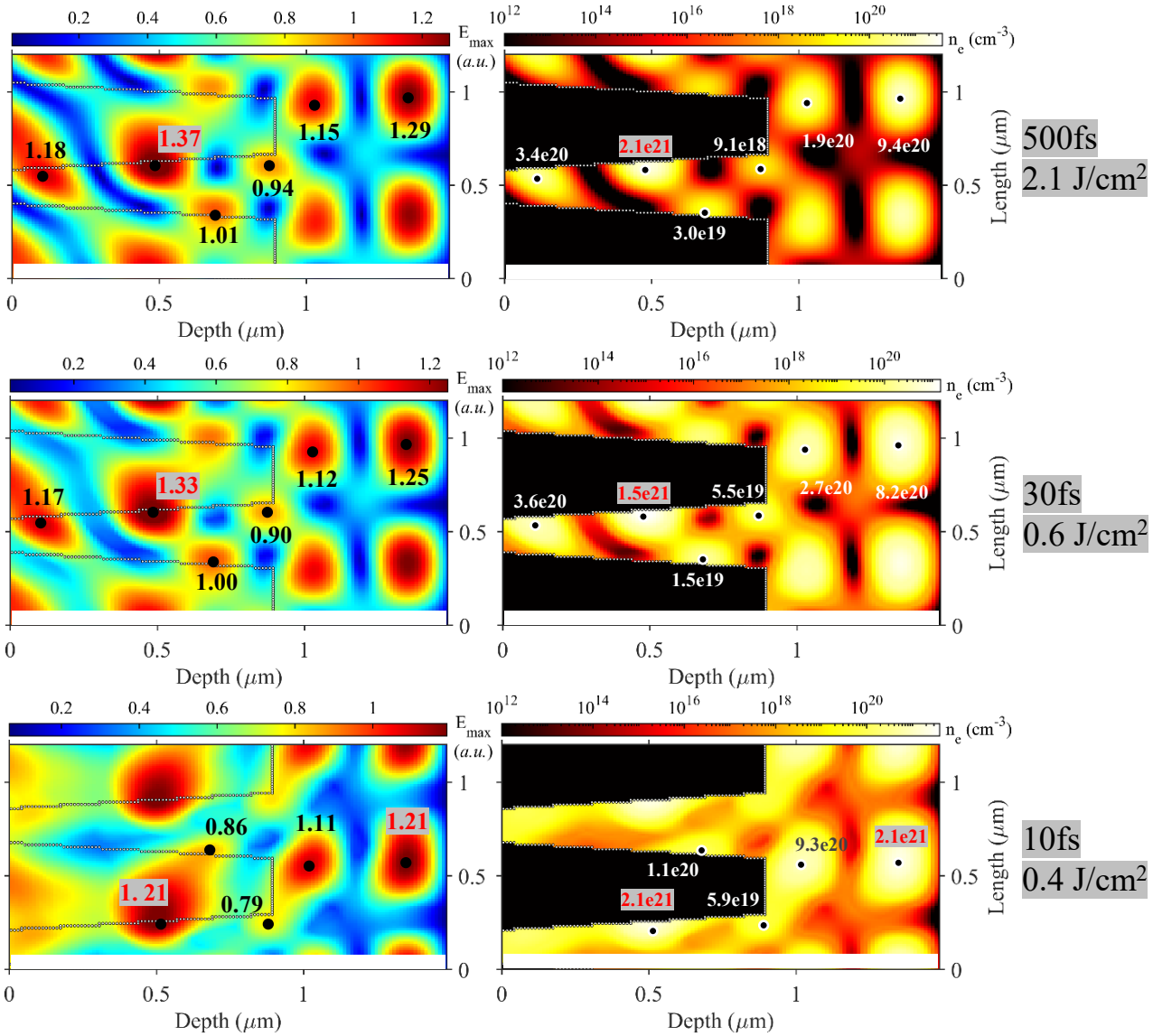


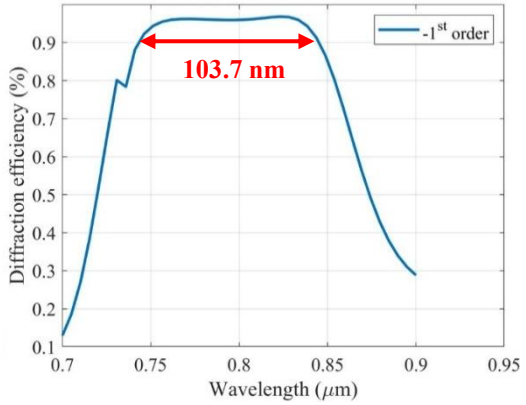
Cross-sectional view of the Grating. Tested LIDT ~0.32 J/cm^2 with a 45 fs pulse. [16]

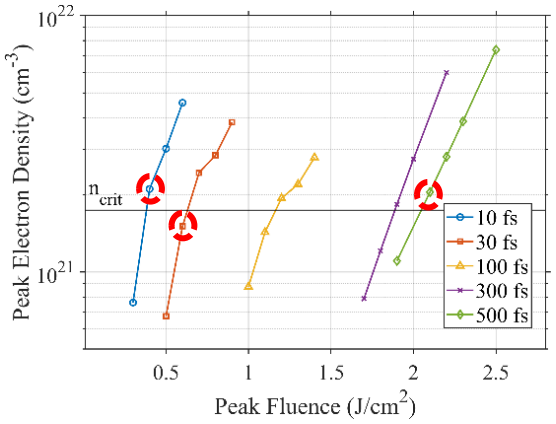
[16] H. Guan, H. Chen, J. Wu, et al., "High-efficiency, broad-bandwidth metal/multilayerdielectric gratings," Opt. Lett. 39, 170–173 (2014).

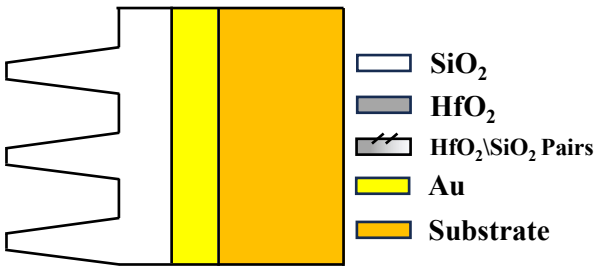
Results Xu

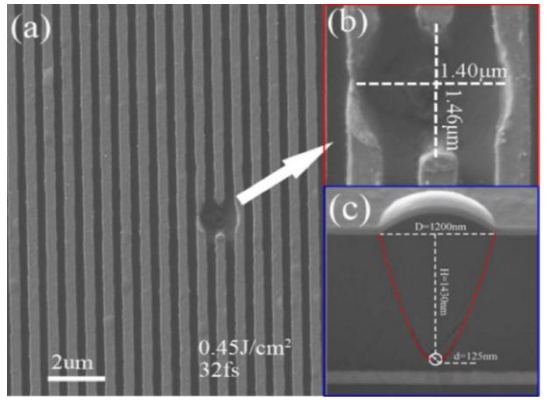
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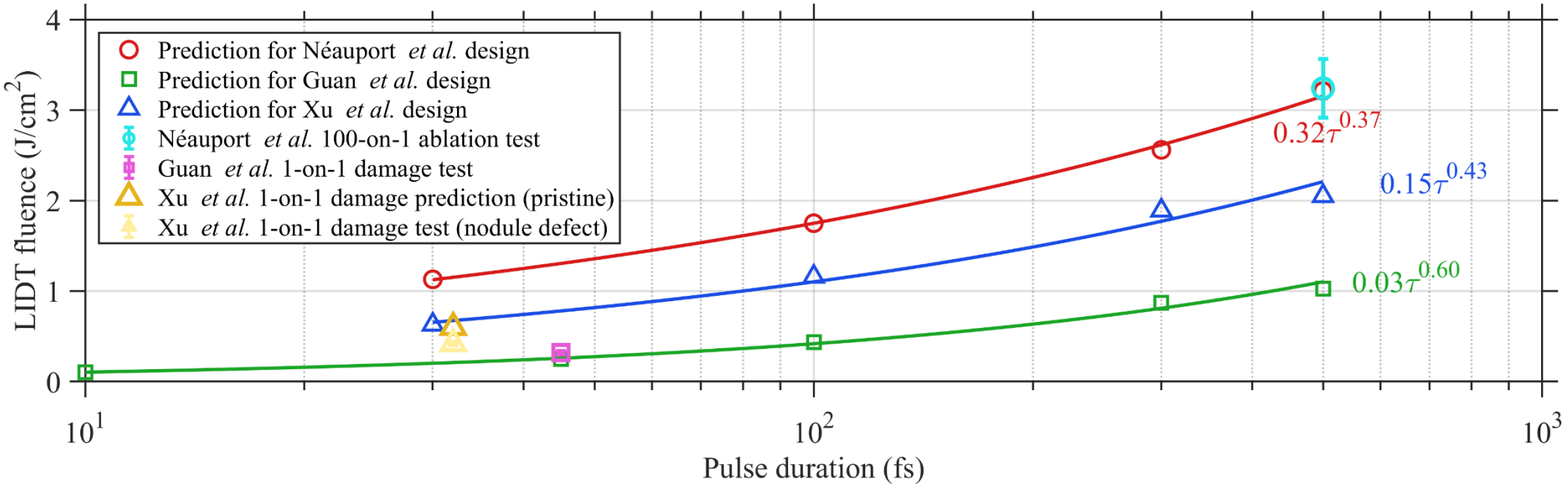




(a) Nodule defects distributed in the MDG; (b) laser induced eruption of the defect, LIDT ~ 0.45 J/cm² with a 32 fs pulse; (c) measured size of nodule defect. [17]

[17] J. Xu, X. Zou, J. Chen, et al., "Metal dielectric gratings with high femtosecond laser damage threshold of twice as much as that of traditional gold gratings," Opt. Lett. 44, 2871–2874 (2019).

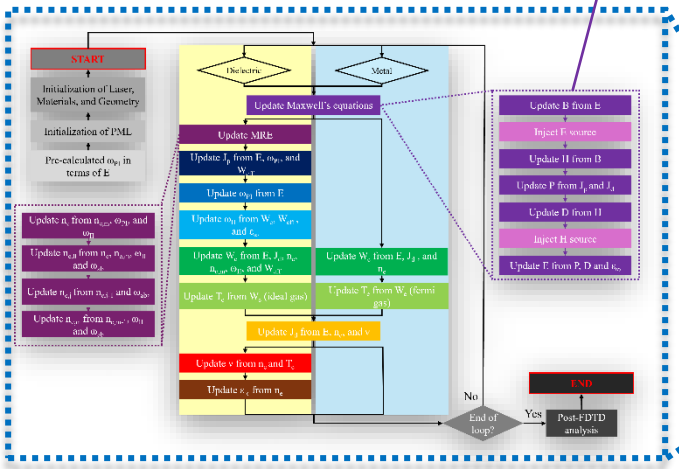
Summary 1

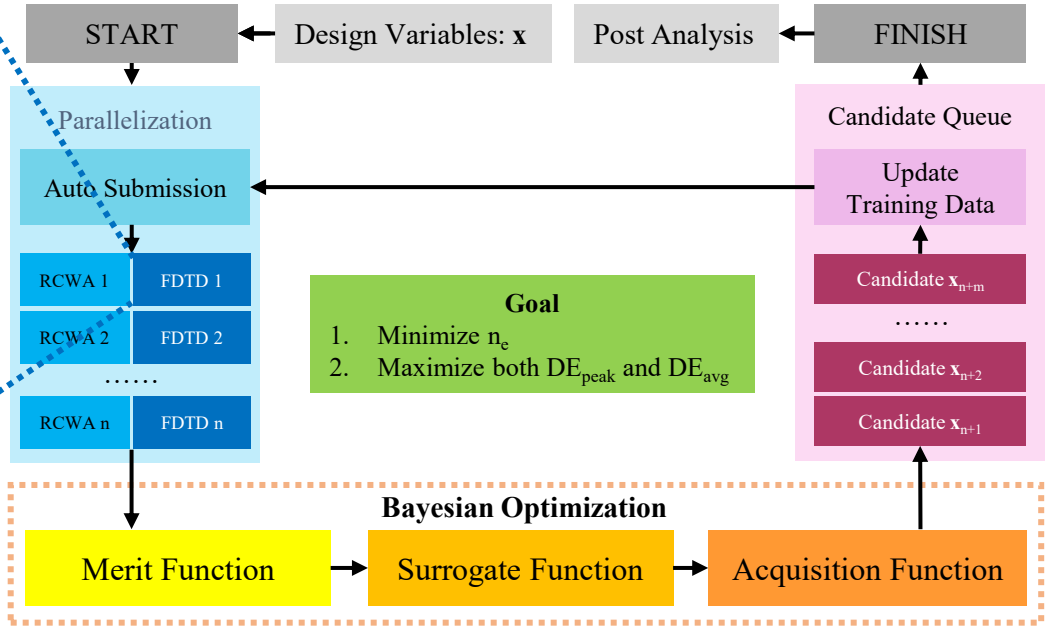


- Good agreement between simulated and experimental LIDTs for three hybrid gratings.
- Scaling exponent α = 0.37–0.60, governed by material properties and field distribution.
- Potential secondary damage sites identified in underlying dielectric layers.
- Higher α achievable without sacrificing diffraction-efficiency bandwidth.
- Future work: Extend TTM to investigate picosecond pulse scaling and evaluate metal layer damage.

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Note: All source are using a super-Gaussian instead of a normal Gaussian to minimize effects from different interaction positions between pulse peak and pillar during optimization.





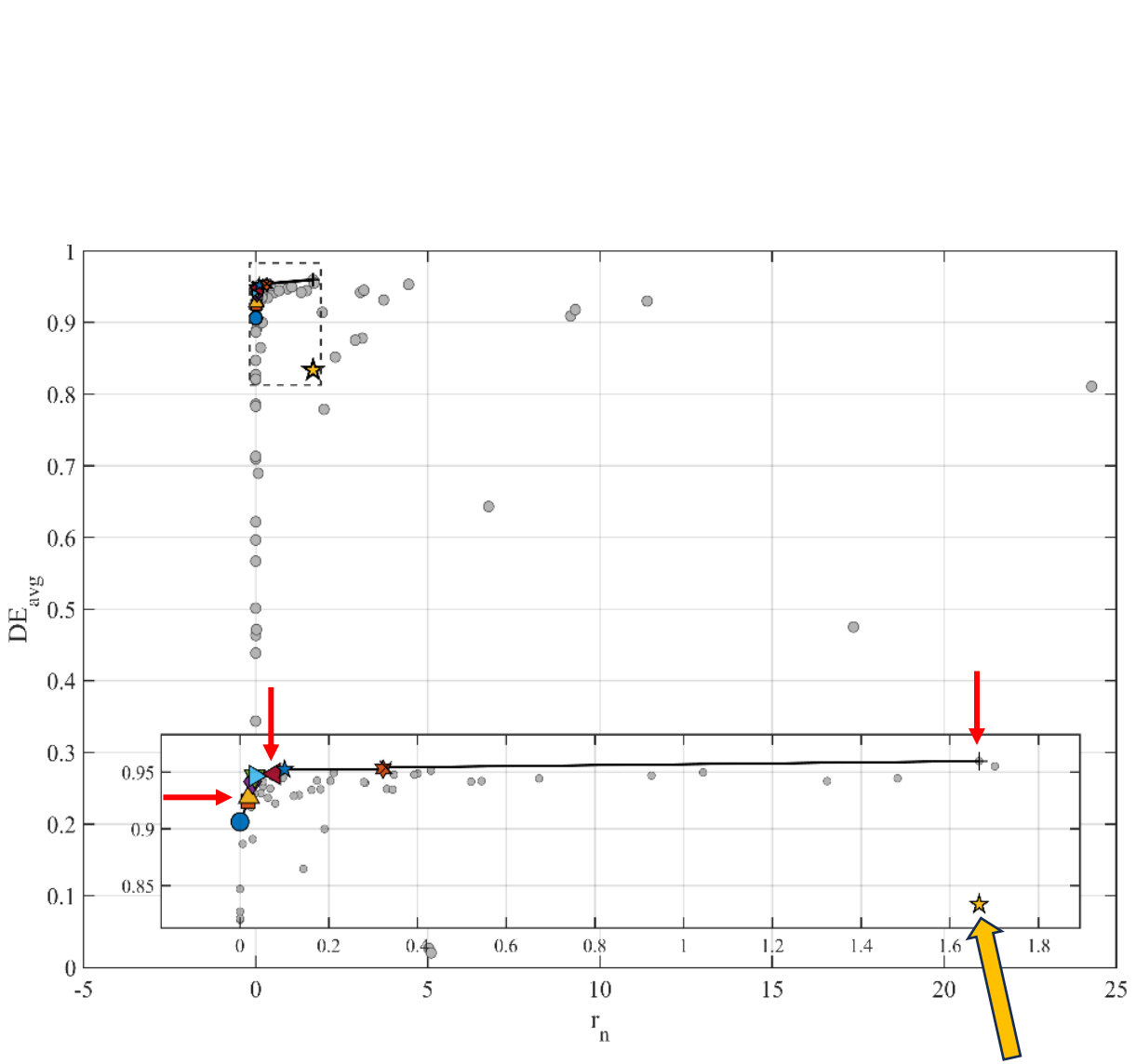
$$\mathrm{MF}(\mathbf{x}) = a\left[\mathrm{DE}_{\mathrm{peak}}(\mathbf{x}) - \mathrm{DE}_{\mathrm{peak},0}\right] + b\left[\mathrm{DE}_{\mathrm{avg}}(\mathbf{x}) - \mathrm{DE}_{\mathrm{avg},0}\right] - c\frac{d_p(\mathbf{x})}{\Delta\omega} - \left[r_n(\mathbf{x}) - 1\right]$$

$$\mathrm{SF}(\mathbf{x}) \sim \mathcal{N}\left(\mu(\mathbf{x}), \sigma^2(\mathbf{x})\right)$$

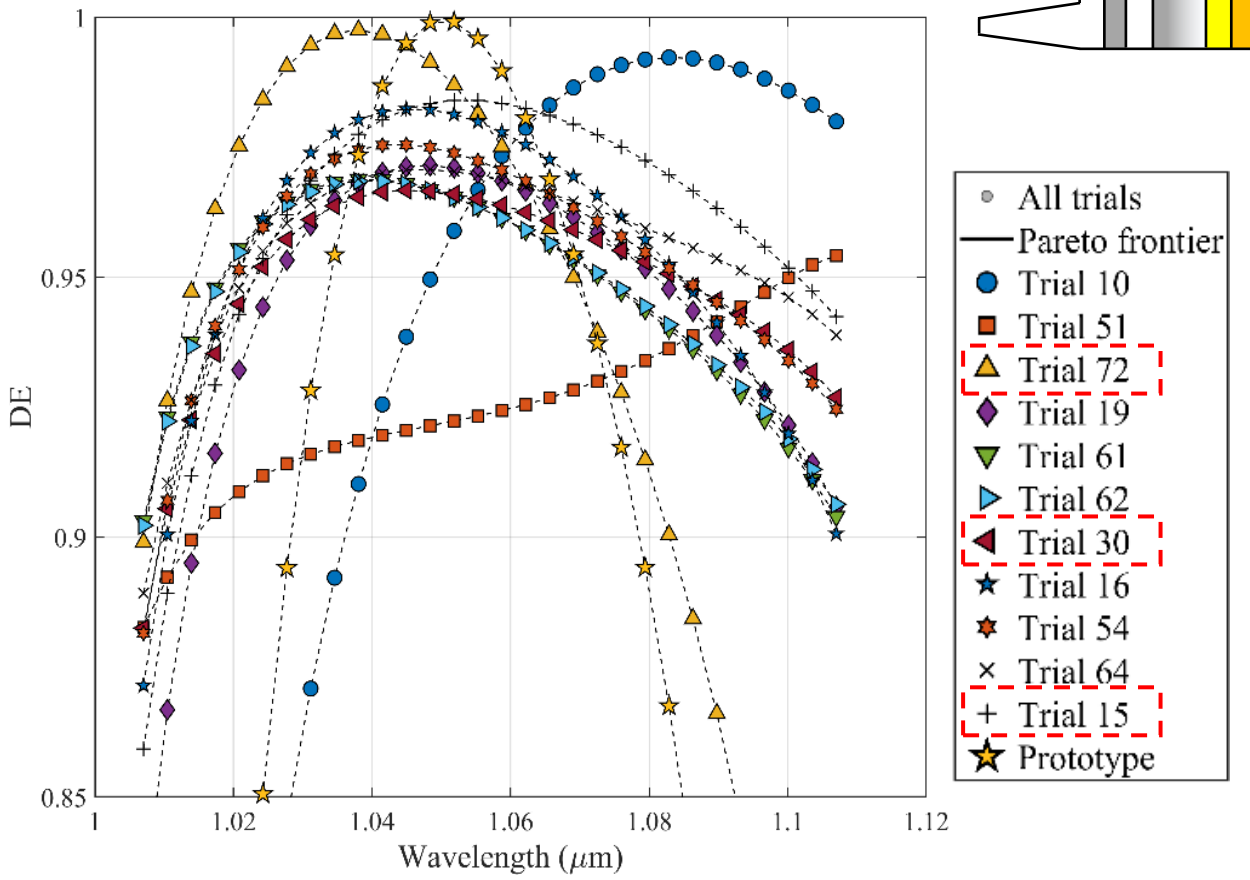
$$\mathrm{AF}(\mathbf{x}) = \mu(\mathbf{x}) + \sqrt{\kappa}\sigma(\mathbf{x})$$

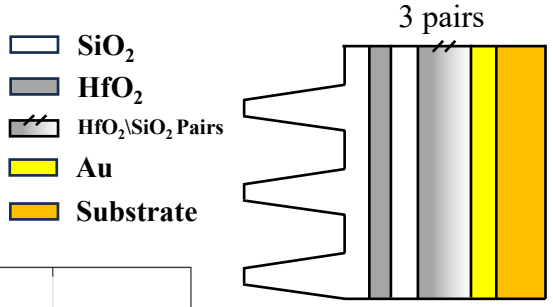
Design	Néauport	Guan	Xu
$DE_{peak,0}$	99%	96%	97%
$DE_{avg,0}$	85%	94%	70%
$\Delta\omega$	100	200	200
a	5	5	5
b	10	10	10
c	0.6	0.6	0.6

Z. Su, et al., "Machine Learning Damage-Aware Optimization of the Dielectric Compression Grating for Ultrafast High-Power Lasers," Manuscript in preparation.

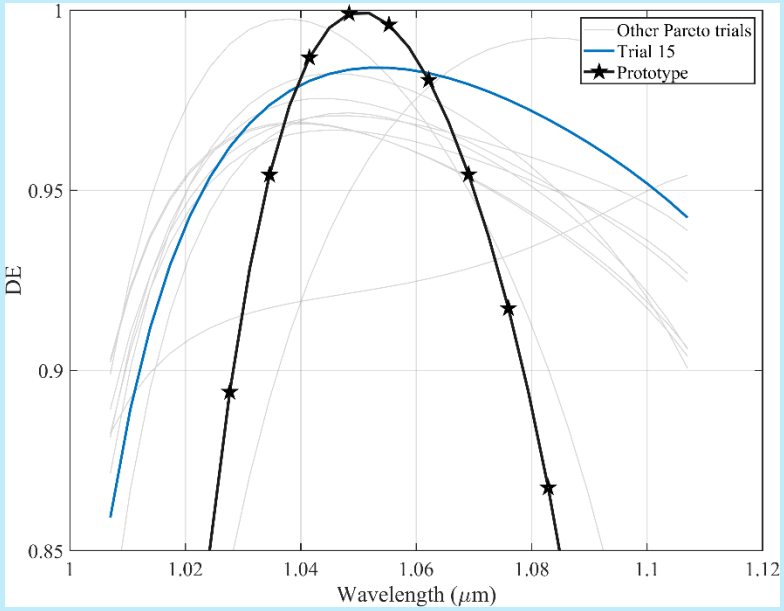

DE_avg
r_n
0.95
0.9
0.85

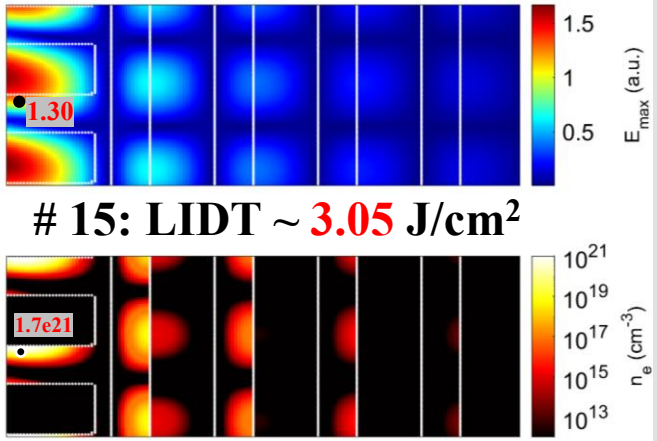
Prototype

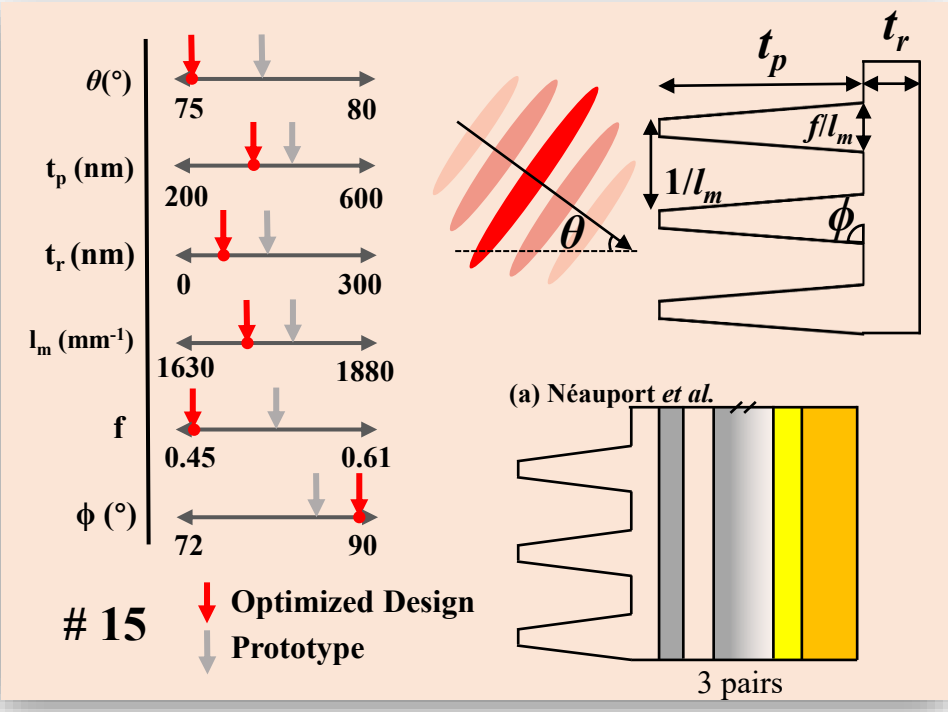

DE
Wavelength (μm)
All trials
Pareto frontier
Trial 10
Trial 51
Trial 72
Trial 19
Trial 61
Trial 62
Trial 30
Trial 16
Trial 54
Trial 64
Trial 15
Prototype


SiO2
HfO2
HfO2\SiO2 Pairs
Au
Substrate
3 pairs

Results Néauport





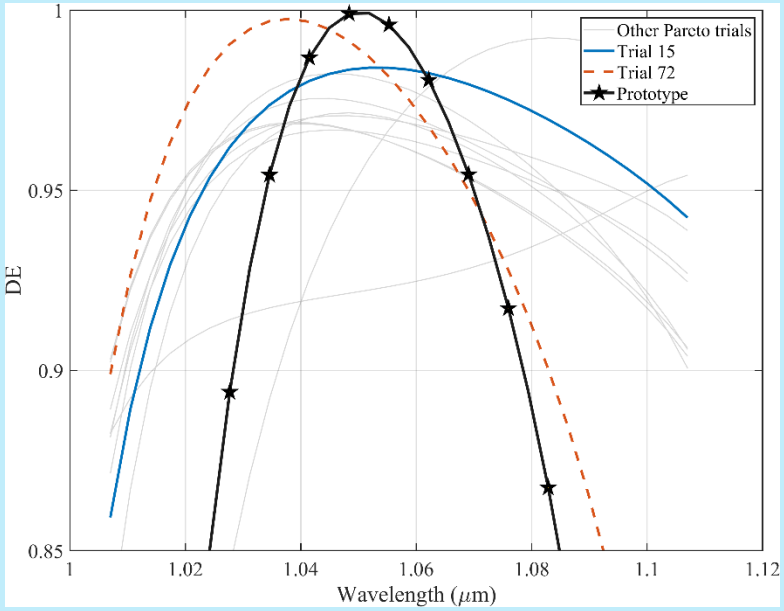


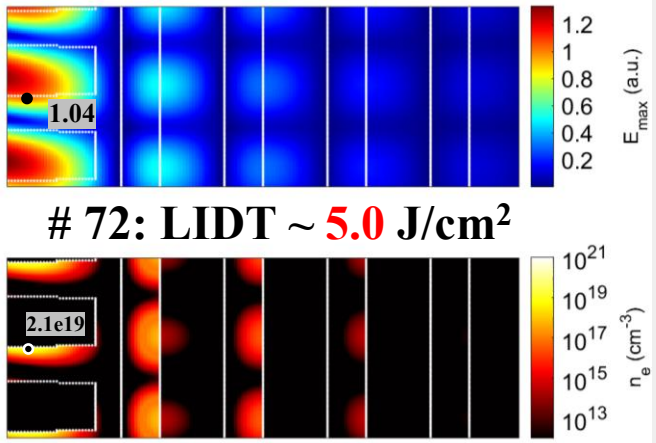
Simulations at 3.2 J/cm²

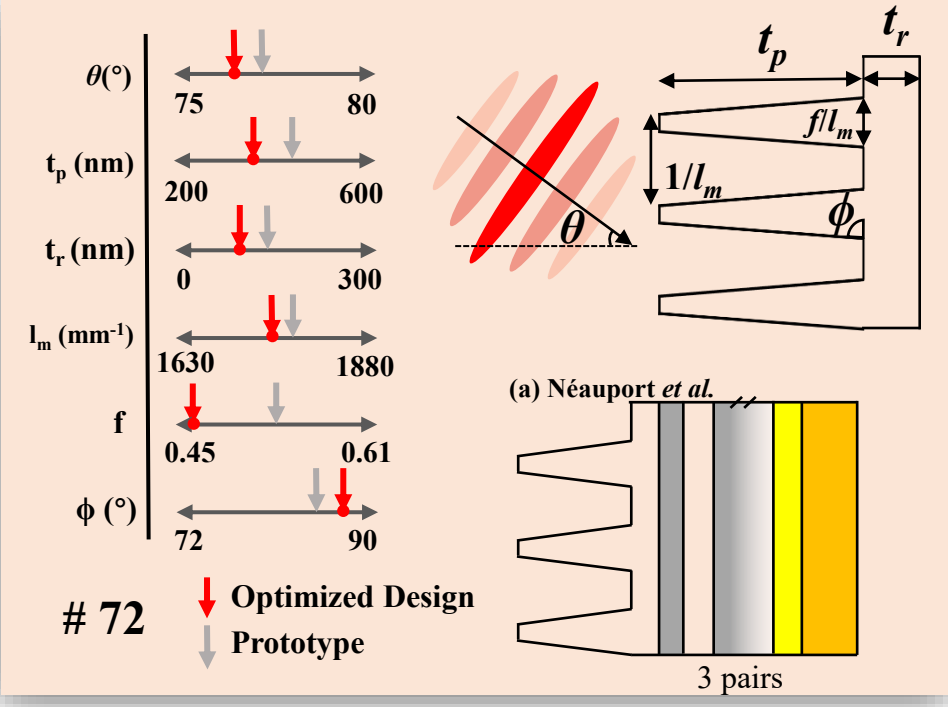
15: Δω ↑ 63nm for DE > 90%

Results Néauport

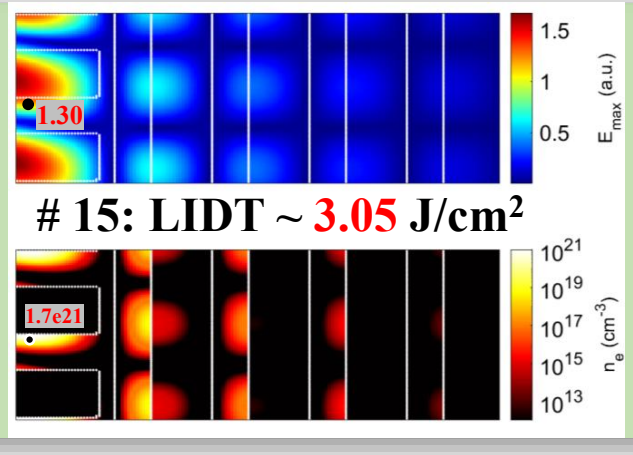
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Simulations at 3.2 J/cm²

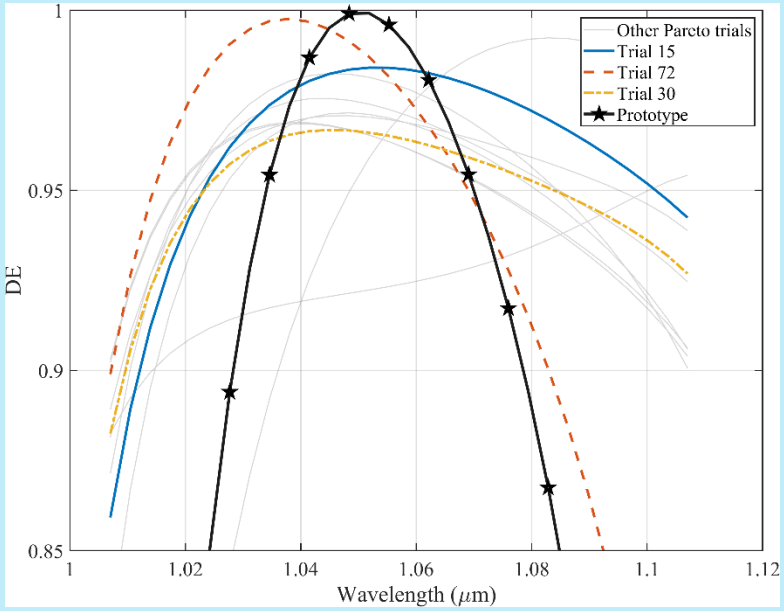


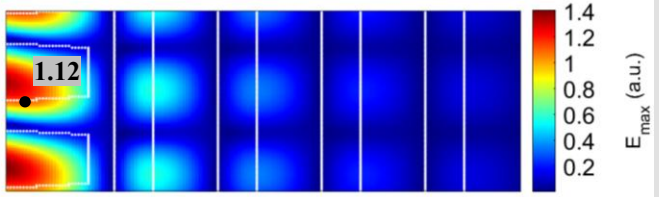
15: Δω ↑ 63nm for DE > 90%

72: Δω ↑ 26nm for DE > 90%
LIDT ↑ 56%

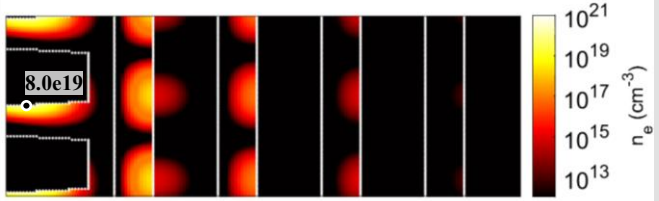
Results Néauport

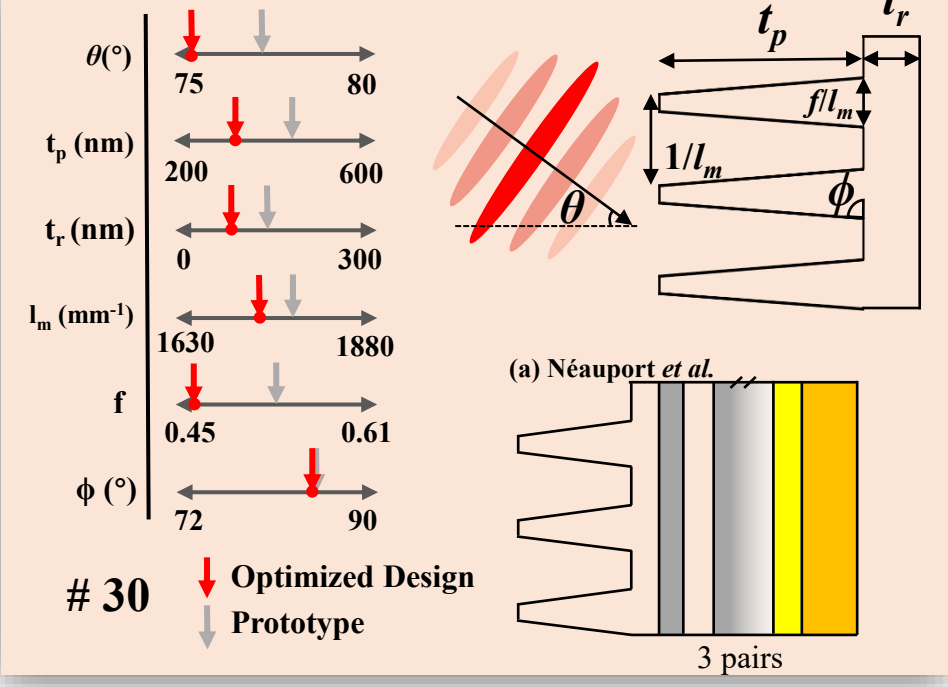
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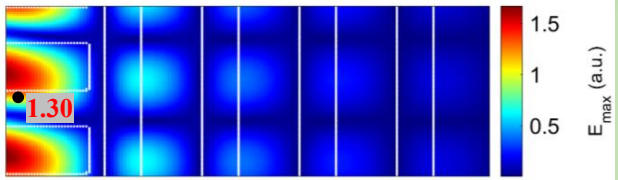


30: LIDT ~ 4.4 J/cm²

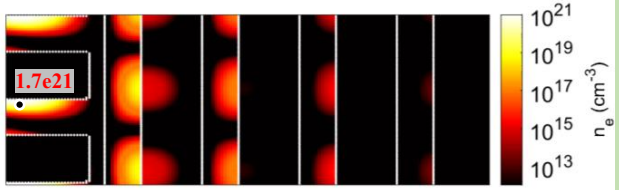




Simulations at 3.2 J/cm²



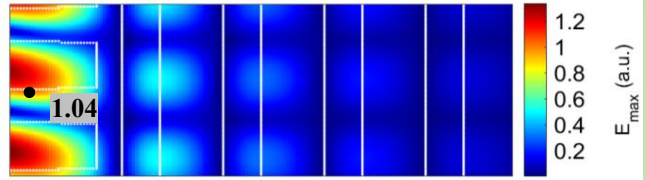
15: LIDT ~ 3.05 J/cm²



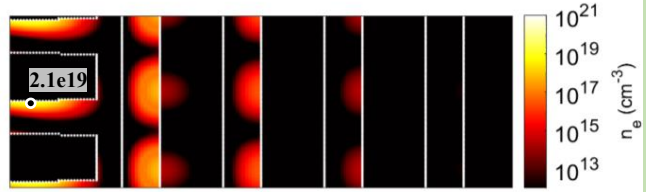
15: Δω ↑ 63nm for DE > 90%

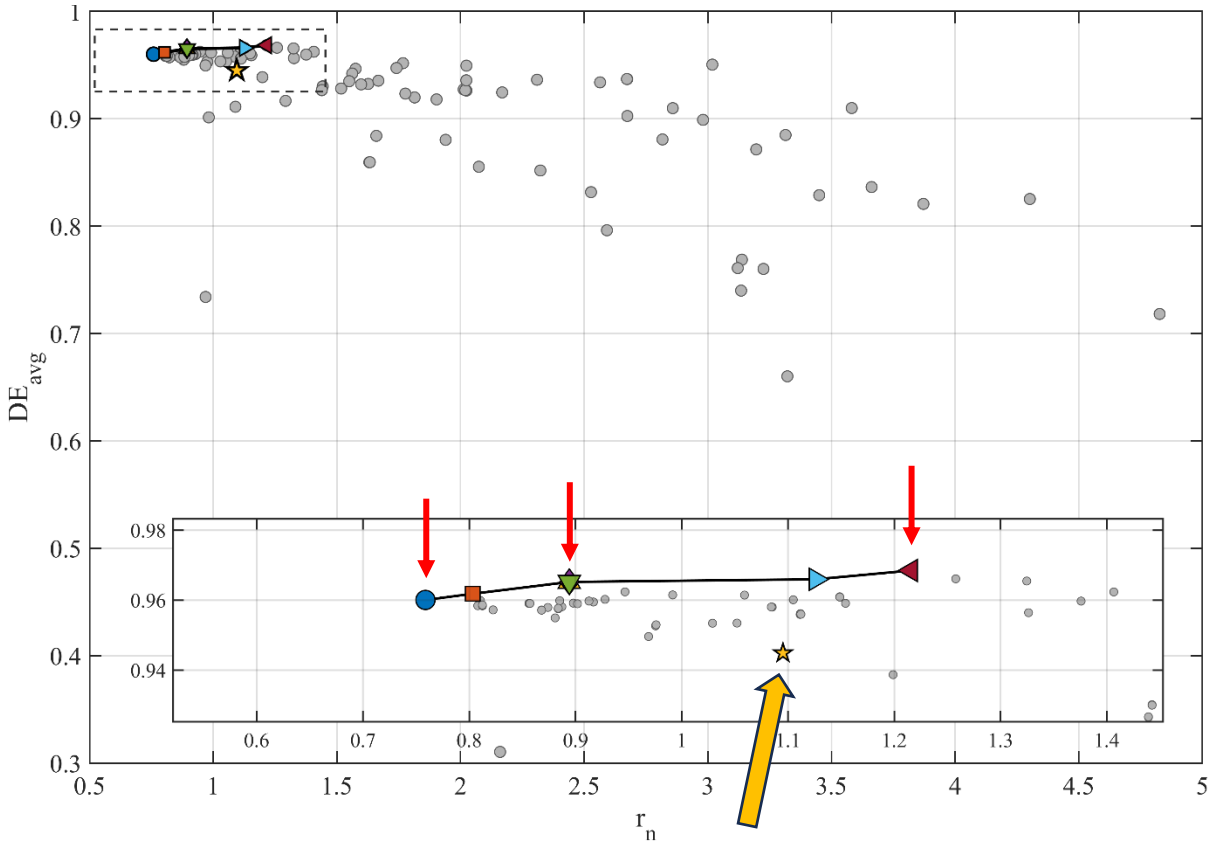
72: Δω ↑ 26nm for DE > 90%
LIDT ↑ 56%

30: Δω ↑ 60nm for DE > 90%
LIDT ↑ 38%

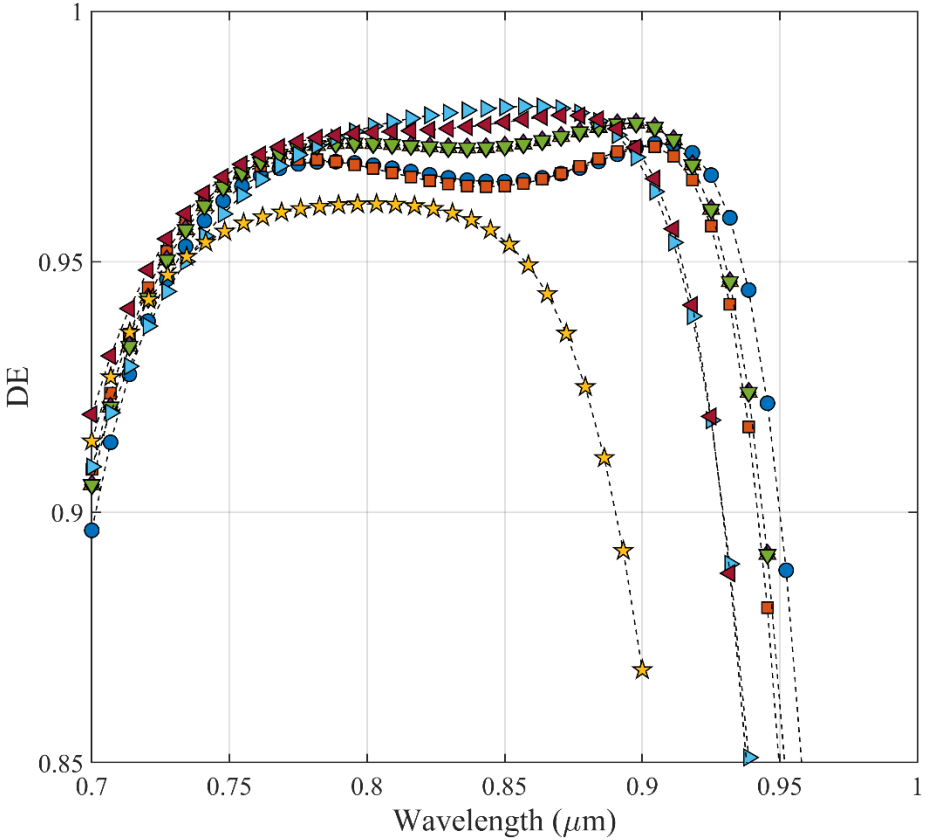


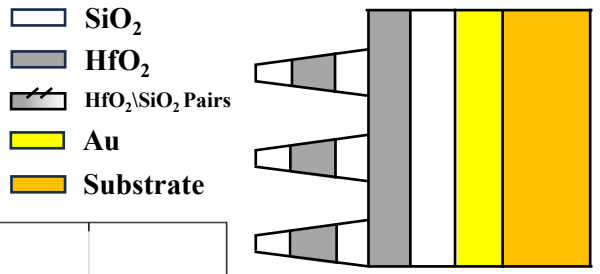
72: LIDT ~ 5.0 J/cm²

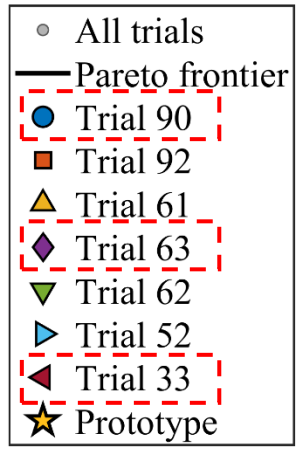



DE_avg
r_n
0.98
0.96
0.94
0.6
0.7
0.8
0.9
1
1.1
1.2
1.3
1.4

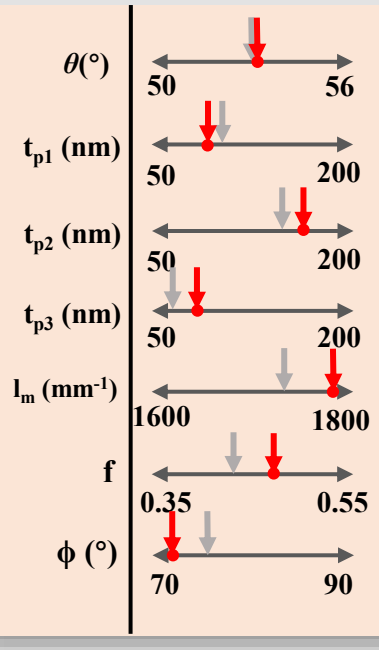
Prototype

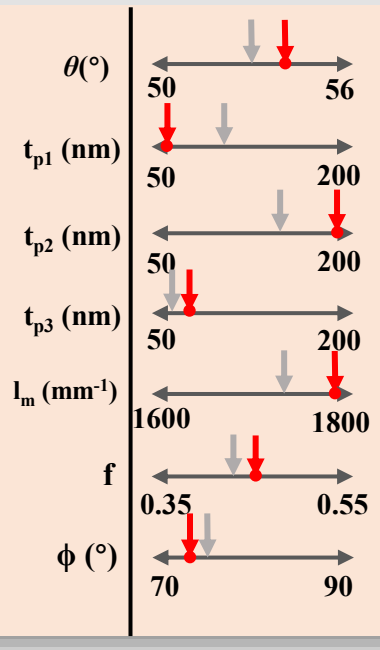

DE
Wavelength (μm)
0.85
0.9
0.95
0.7
0.75
0.8
0.85
0.9
0.95
1

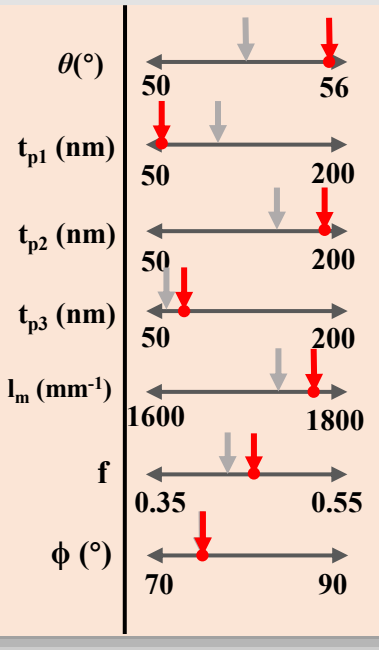

SiO2
HfO2
HfO2\SiO2 Pairs
Au
Substrate


All trials
Pareto frontier
Trial 90
Trial 92
Trial 61
Trial 63
Trial 62
Trial 52
Trial 33
Prototype

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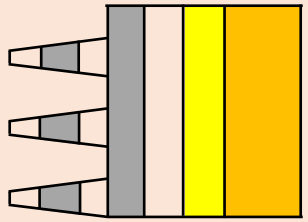


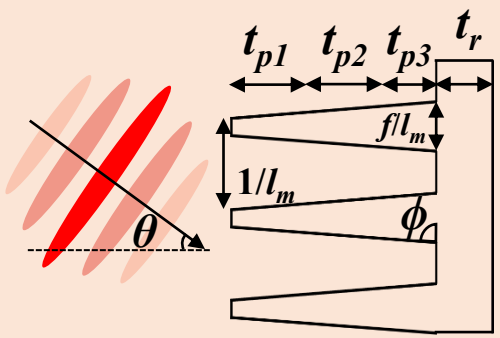


Optimized Design

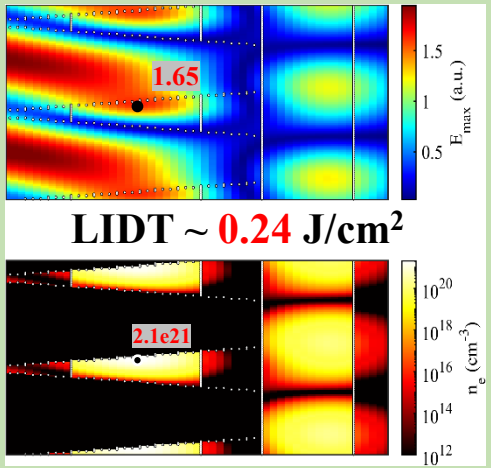
Prototype

(b) Guan *et al.*

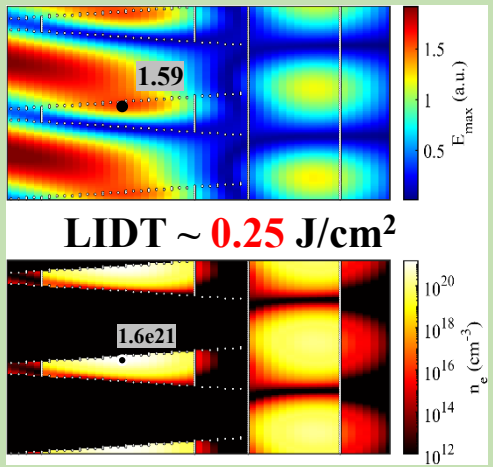




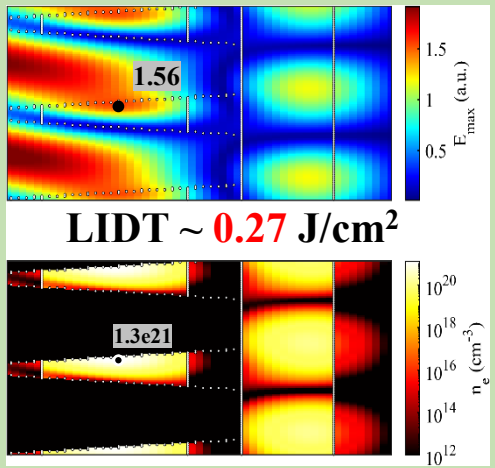
33 **# 63** **# 90**



LIDT ~ 0.24 J/cm²

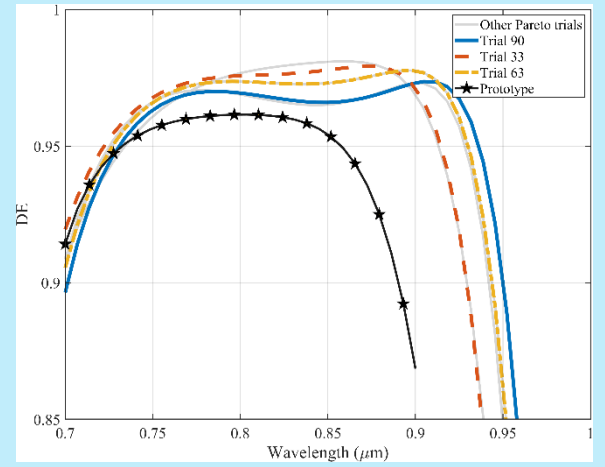


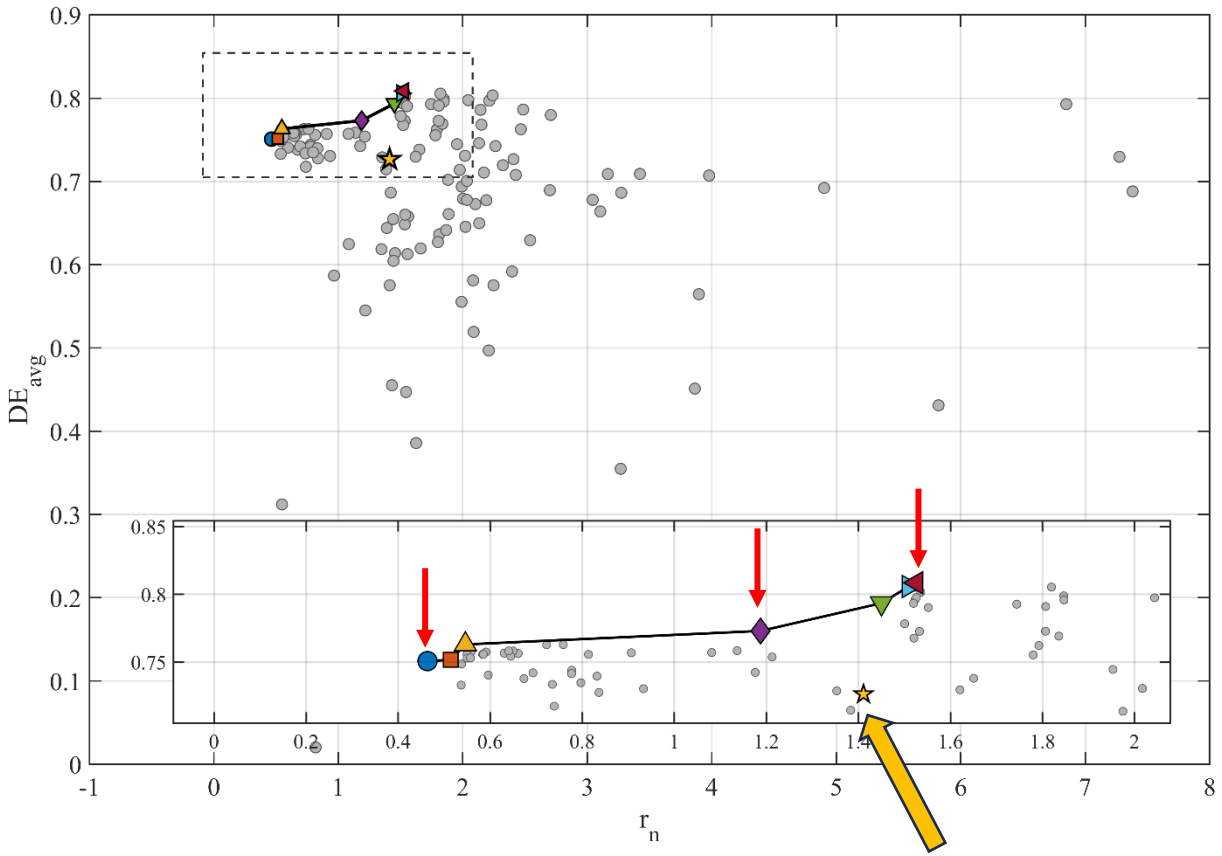
LIDT ~ 0.25 J/cm²



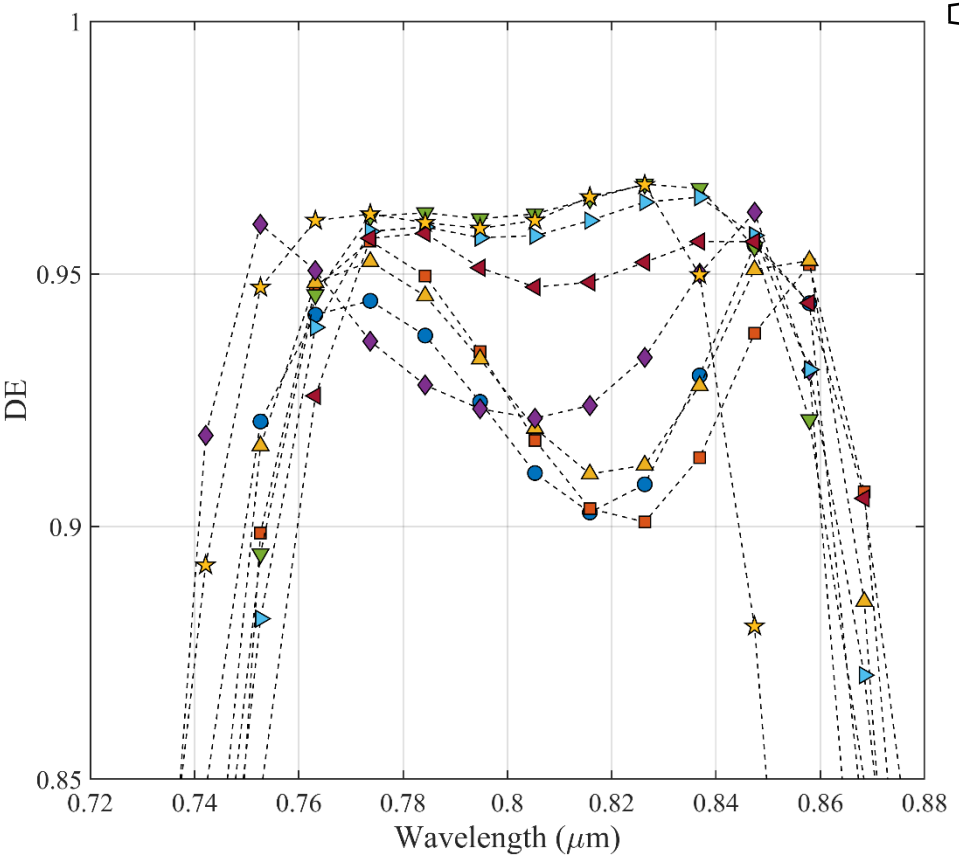
LIDT ~ 0.27 J/cm²

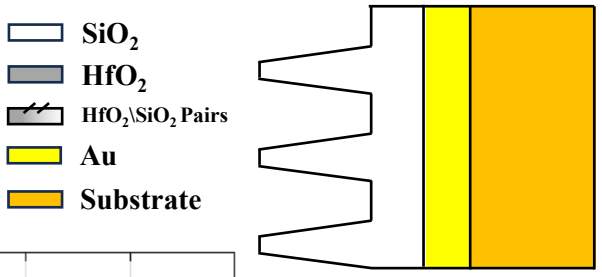
Simulations at 0.25 J/cm²

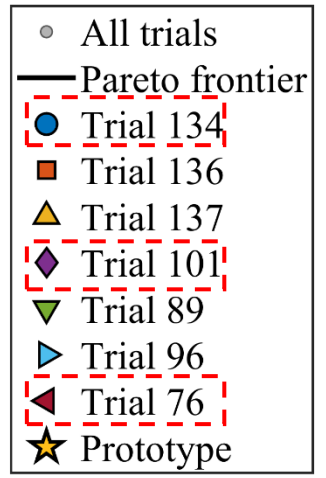



DE_avg
r_n

Prototype

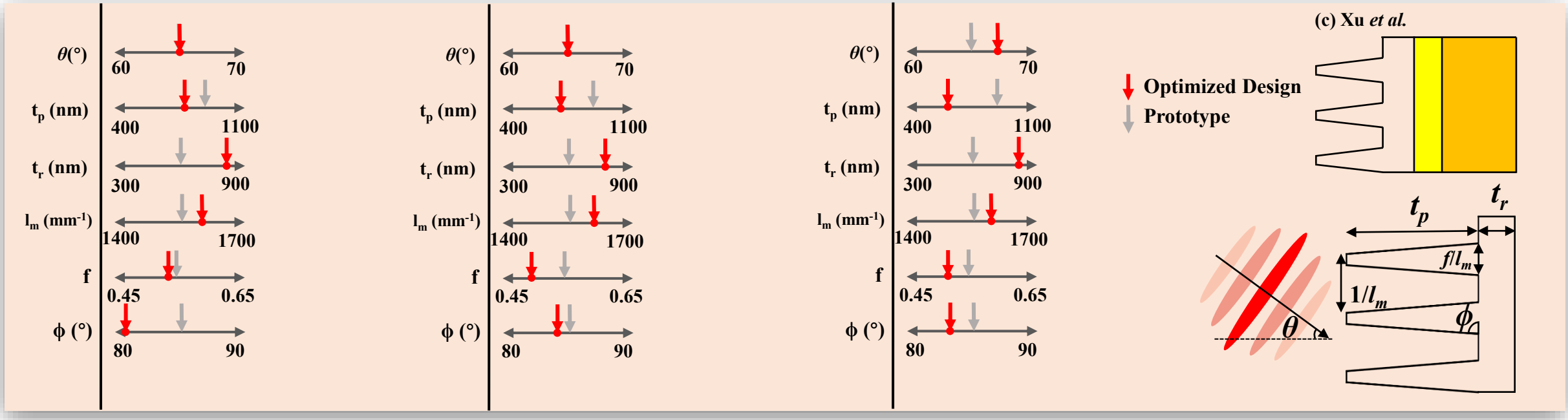

DE
Wavelength (μm)


SiO2
HfO2
HfO2\SiO2 Pairs
Au
Substrate

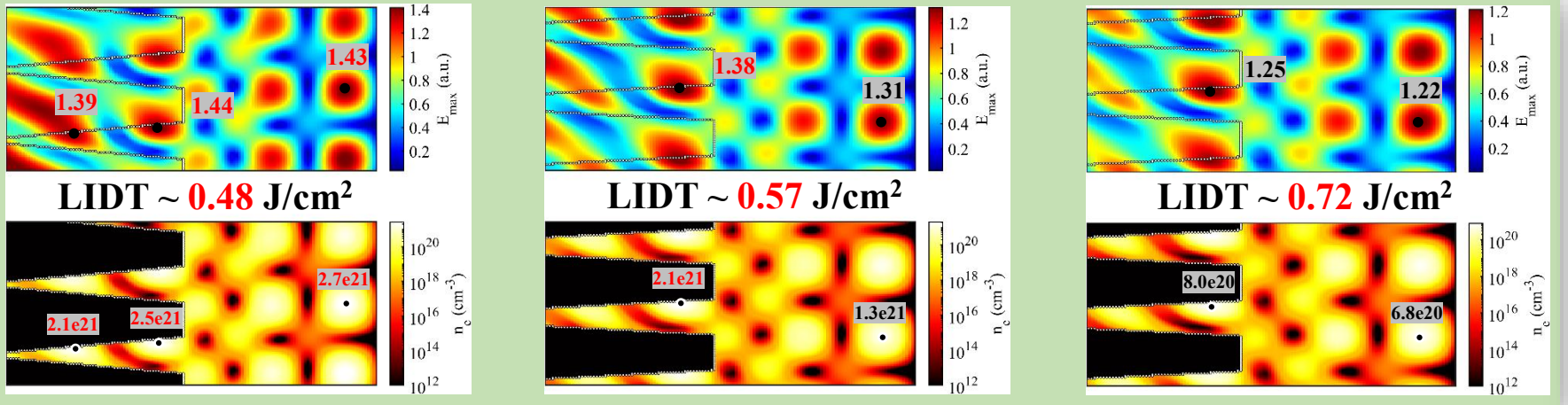

All trials
Pareto frontier
Trial 134
Trial 136
Trial 137
Trial 101
Trial 89
Trial 96
Trial 76
Prototype

Results Xu

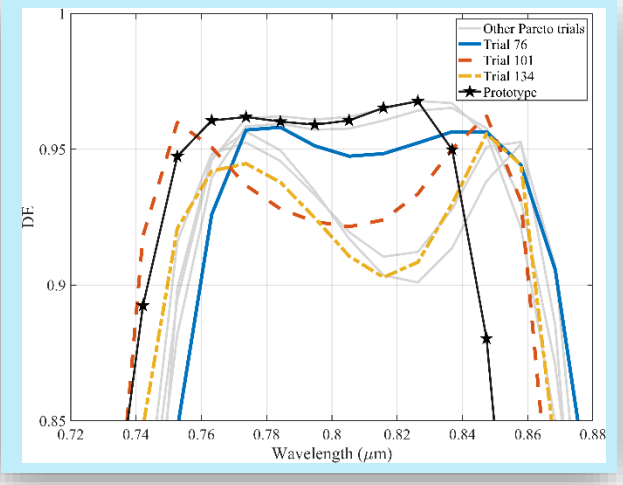
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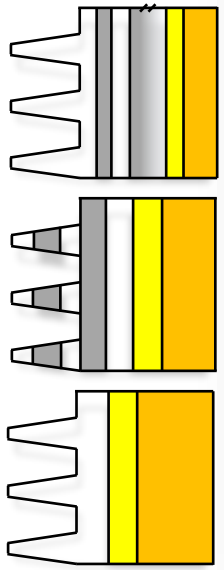
76 # 101 # 134



Simulations at 0.6 J/cm²



Summary 2



Note: All simulations used a super-Gaussian source for calculating LIDT.

	Trial	LIDT (J/cm^2)		Δω (nm) for DE>90%		DE at λ_0	DE_{peak}
Néauport	Prototype	3.05		49		99.75%	99.91% (1052 nm)
	# 15	3.05	~	112	↑ 130%	**98.41%**	**98.41%** (1053 nm)
	# 30	4.4	↑ 38%	109	↑ 124%	**96.53%**	**96.67%** (1045 nm)
	# 72	5	↑ 56%	75	↑ 54%	**98.45%**	**99.76%** (1038 nm)
Guan	Prototype	0.25		193		96.15%	96.15% (800 nm)
	# 33	0.24	~	232	↑ 20%	**97.58%**	**97.92%** (870 nm)
	# 63	0.25	~	246	↑ 27%	**97.36%**	**97.77%** (898 nm)
	# 90	0.27	~	245	↑ 27%	**96.92%**	**97.36%** (905 nm)
Xu	Prototype	0.52		104		95.98%	96.76% (826 nm)
	# 76	0.48	↓ 8%	108	~	**94.93%**	**95.70%** (774 nm)
	# 101	0.57	↑ 10%	118	↑ 13%	**92.24%**	**96.23%** (847 nm)
	# 134	0.72	↑ 38%	116	↑ 12%	**91.76%**	**95.57%** (847 nm)

- All designs have space to improve with some trade-offs.
- Successfully coupled the damage-aware capability into grating optimization.
- In most of cases, a lower duty cycle (between 0.4~0.5) may increase the LIDT.
- Future work: Try different merit functions to achieve different optimization goals

Thank you

For more questions, please contact:
su.775@osu.edu

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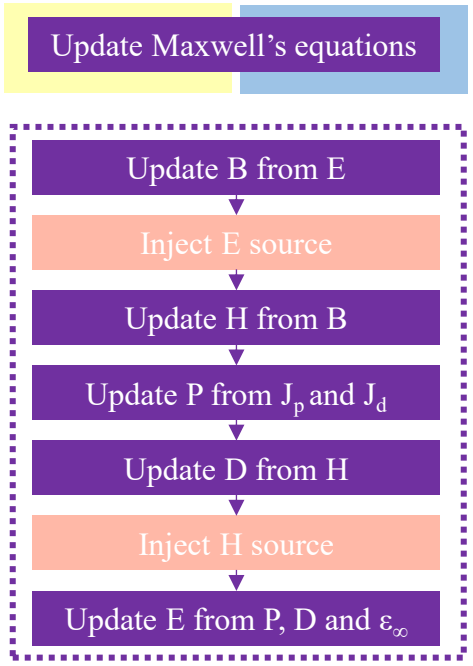


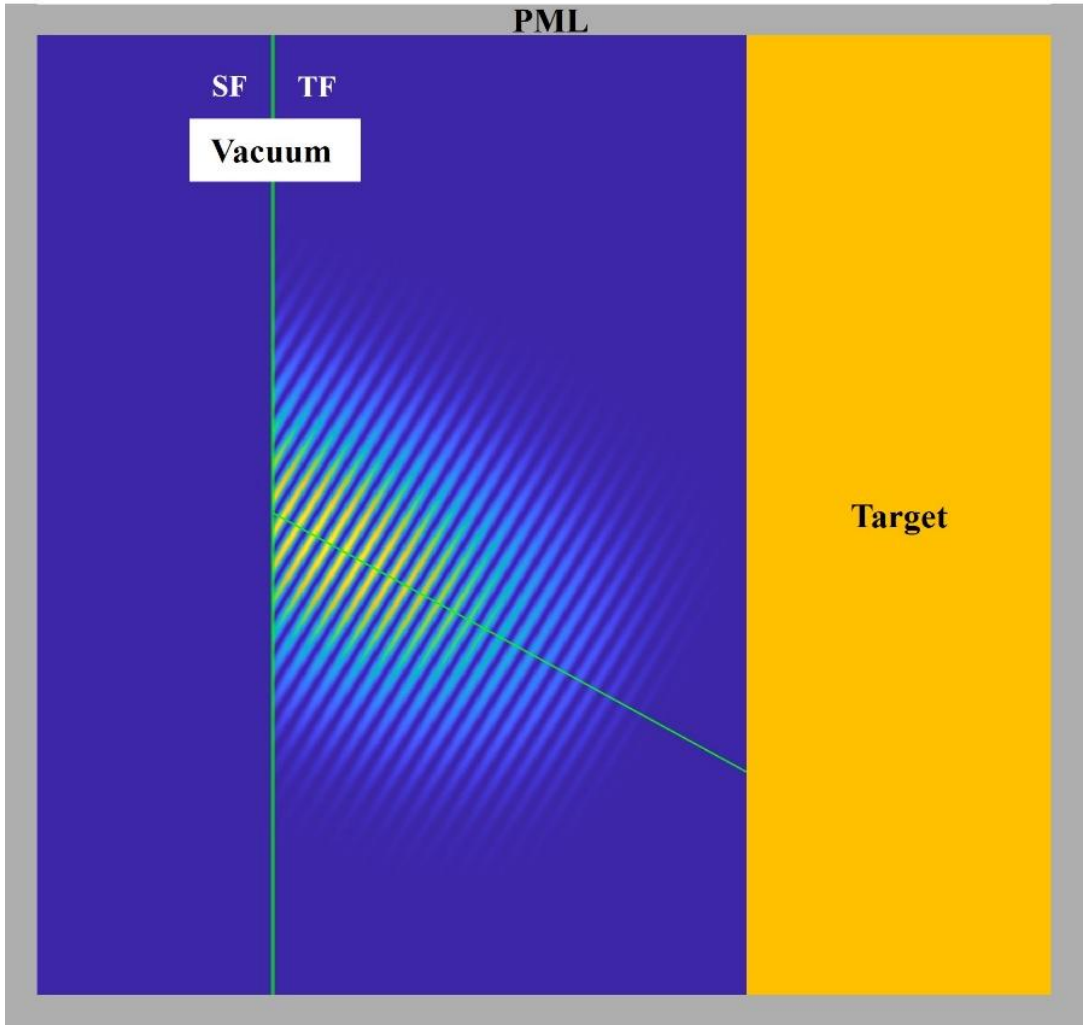


Bibliography

[1] M. F. Kling, et al., Basic Research Needs Workshop on Laser Technology Report of the Department of Energy Office of Science Workshop August 15-17, 2023 (U.S. Department of Energy Office of Science, January 5, 2024), iii, 4, 44, 49-51, 54, etc.
[2] L Kiani., et al., "High Average Power Ultrafast Laser Technologies for Driving Future Advanced Accelerators" (technical report), Journal of Instrumentation 18, no. 08 (2023): 9
[3] Lawrence Livermore National Laboratory, Advanced Optical Technologies. Available at: https://lasers.llnl.gov/science/photon-science/advanced-optics
[4] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)
[5] S. Zhang, Z. Su, C. S. Menoni, and E. A. Chowdhury, "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings," Opt. Lett., vol. 48, no. 5, pp. 1212–1215, Mar. 2023.
[6] Ziyao Su, et al., "Keldysh ionization based-FDTD modeling of laser-induced damage threshold of MLD-IBS compression gratings for petawatt 2 μm laser systems", Proc. SPIE PC13729, Laser-Induced Damage in Optical Materials 2025, PC1372907 (19 Dec 2025);
[7] M. Y. Noor, et al., "High Damage Threshold Multilayer Dielectric Grating Enabled for 2 μm High-Energy CPA Systems," Manuscript in preparation.
[8] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)
[9] M. Y. Noor, et al., "Pulse duration dependent effects of ultrafast laser induced damage on a 1030 nm multi-layer dielectric mirror for high repetition rate, high average power laser systems," Opt. Mater. Express 15, 212-228 (2025)
[10] Z. Su and E. A. Chowdhury, "Ultrafast laser interaction with dielectric–metal hybrid coatings: a two-dimensional finite-difference time-domain modeling study," High Power Laser Science and Engineering, vol. 14, p. e78, 2026.
[11] Z. Su and E. A. Chowdhury, "Pulse-duration scaling of ultrafast laser-induced damage threshold in hybrid gratings," Opt. Lett. 51, 4236-4239 (2026)
[12] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[13] L. V. Keldysh, "Kinetic theory of impact ionization in semiconductors," SOVIET PHYSICS JETP, vol. 37, no. 10, 1960.
[14] B. Rethfeld, "Unified Model for the Free-Electron Avalanche in Laser-Irradiated Dielectrics," Physical Review Letters 92, 187401 (2004).
[15] J. Néauport, N. Bonod, S. Hocquet, et al., "Mixed metal dielectric gratings for pulse compression," Opt. Express 18, 23776–23783 (2010).
[16] H. Guan, H. Chen, J. Wu, et al., "High-efficiency, broad-bandwidth metal/multilayerdielectric gratings," Opt. Lett. 39, 170–173 (2014).
[17] J. Xu, X. Zou, J. Chen, et al., "Metal dielectric gratings with high femtosecond laser damage threshold of twice as much as that of traditional gold gratings," Opt. Lett. 44, 2871–2874 (2019).
[18] P. Balling and J. Schou, "Femtosecond-laser ablation dynamics of dielectrics: Basics and applications for thin films," Reports on Progress in Physics, vol. 76, no. 3, p. 036 502, Feb.2013. DOI: 10.1088/0034-4885/76/3/036502

Back-up

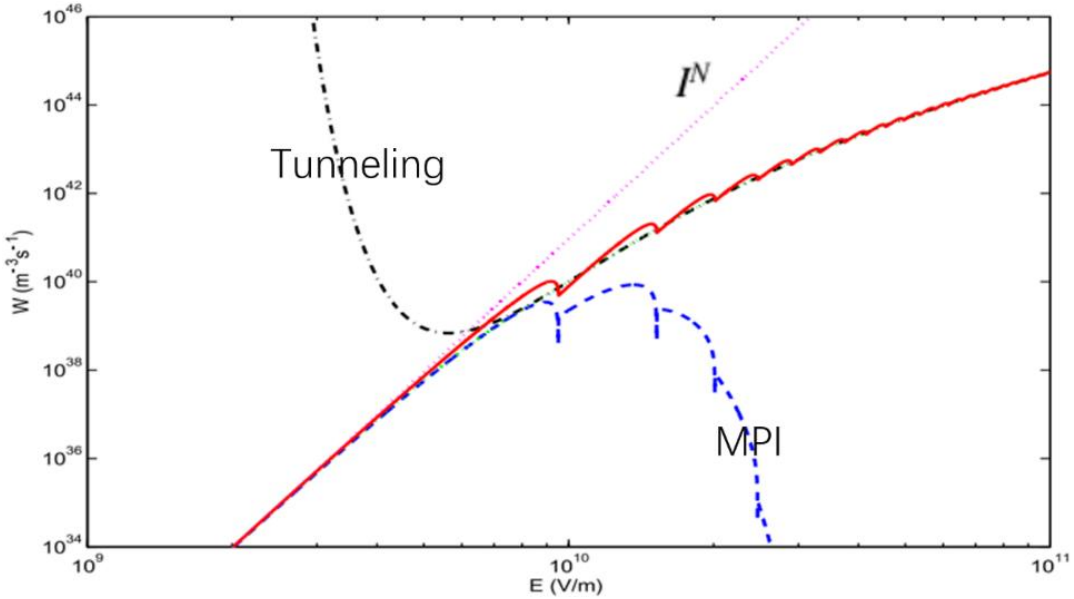




- **TFSF enables to ~ps scale**

[12] Allen Taflove, Susan C. Hagness, "Computational Electrodynamics The Finite-Difference Time-Domain Method" (2000)

Photoionization[12]

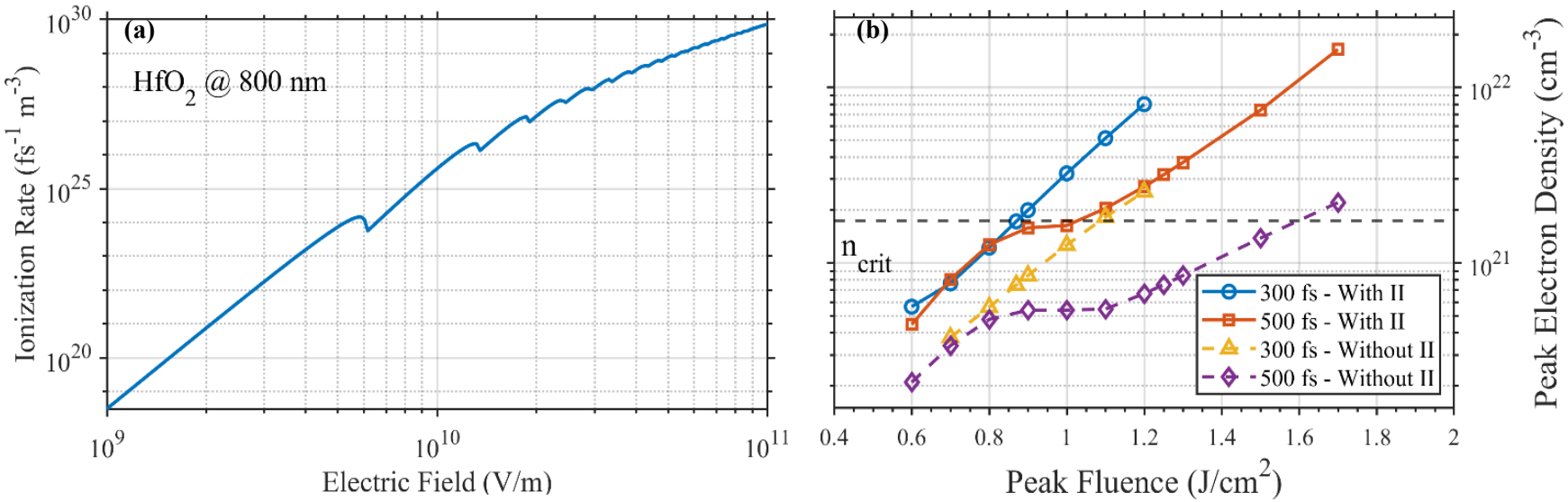


Strong Field Ionization Rates Based on Different Models (wavelength 800 nm, bandgap 6.5 eV) [3]

$$\gamma = \frac{\omega\sqrt{m_e W_0}}{eE}$$

W_0: Ionization energy
E: External electric field
ω : Laser frequency

[12] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[18] P. Balling and J. Schou, "Femtosecond-laser ablation dynamics of dielectrics: Basics and applications for thin films," Reports on Progress in Physics, vol. 76, no. 3, p. 036 502, Feb.2013. DOI: 10.1088/0034-4885/76/3/036502



(a) HfO2 photoionization rate at 800 nm for the Guan etal. design. (b) Peak electron density versus fluence at 300 and500 fs, with and without impact ionization. [11]

[11] Z. Su and E. A. Chowdhury, "Pulse-duration scaling of ultrafast laser-induced damage threshold in hybrid gratings," Opt. Lett. 51, 4236-4239 (2026)